

Intrinsic Uniqueness and Reconstruction Across Mathematical Presentations

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15 September 2026

Abstract

The complete mathematical presentations established in this work are not intrinsically distinct objects. They are exact presentations of the same mathematical object Σ .

In a placed coordinate, Σ is represented by

$$\Sigma_P(r) = \log\left(\frac{\mu^2(1 + \gamma r)}{\mu + \gamma}\right) - \mu r,$$

with placement recoverable exactly from local differential data. Removing placement exposes the intrinsic decomposition

$$H(t) = \log t - t + 1, \quad I(t) = t - 1 - \log t, \quad p(t) = te^{-t}.$$

The central result is a global reconstruction theorem. Every complete identifying presentation established here reconstructs Σ uniquely within its stated candidate class, and Σ reconstructs that presentation exactly. Consequently, the apparent separateness of these complete presentations is representational rather than mathematical.

The reconstruction is proved across differential, group, convex, probabilistic, operator-spectral, formal-series, topological, matrix-geometric, spatial, and arithmetic presentations. Exact inverse maps and uniqueness theorems are accompanied by explicit countermodels showing where incomplete data cease to identify Σ . A relative irredundancy theorem separates intrinsic structure, placement, and externally retained context.

The complete development has been formalized in Lean 4 with no admitted proofs.

1 The intrinsic object and placement

The mathematical object Σ is given first in a placed coordinate. Its placement is recovered exactly, and removing it exposes the intrinsic decomposition used throughout the paper.

Definition 1.1 (The placed presentation of Σ). For $\mu > 0$, $0 < a < 1$, put $P = (\mu, a)$, $\gamma = \mu/a$, $b = -a/\mu$, and $r_* = (1 - a)/\mu$. The placed presentation is

$$\Sigma_P(r) = \log\left(\frac{\mu^2(1 + \gamma r)}{\mu + \gamma}\right) - \mu r, \quad r > b.$$

The intrinsic and placed coordinates, their units, and the parameter marks are part of these definitions.

Theorem 1.2 (Exact placed reconstruction). *The map $t = \mu r + a$ bijects (b, ∞) with $(0, \infty)$ and gives*

$$\Sigma_P(r) = \log t - t + \log \frac{\mu}{1 + a} + a.$$

Inside the placed family, at every point of that domain, placement is recovered by

$$\mu = \sqrt{-\Sigma_P'' - \Sigma_P'}, \quad \gamma = \frac{\sqrt{-\Sigma_P''}}{1 - r\sqrt{-\Sigma_P''}}, \quad a = \frac{\mu}{\gamma}.$$

Removing this recovered placement and the value at r_* exposes the placement-free intrinsic form

$$H(t) := \Sigma_P\left(\frac{t - a}{\mu}\right) - \Sigma_P(r_*) = \log t - t + 1, \quad t > 0.$$

Thus

$$\Sigma_P(r) = H(\mu r + a) + \log \frac{\mu}{1 + a} + a - 1. \quad (1)$$

Its convex negative $I(t) = t - 1 - \log t$ and normalized density $p(t) = te^{-t}$ complete the intrinsic decomposition (H, I, p) of Σ , with the exact identities

$$I = -H, \quad p = e^{H-1}, \quad H = 1 + \log p. \quad (2)$$

The closure satisfies $\Sigma_P'(r_*) = 0$, $\Sigma_P''(r_*) = -\mu^2$. The density p has mass one on $(0, \infty)$, whereas

$$e^{\Sigma_P(r)} dr = \frac{p(t) dt}{(1 + a)e^{-a}}.$$

Consequently e^{Σ_P} has mass one on $r \geq 0$ and mass $e^a/(1 + a)$ on its whole analytic domain. The presentation Σ_P reconstructs both its intrinsic decomposition (H, I, p) and its placement P exactly; conversely, (H, I, p) together with P reconstructs Σ_P exactly.

Proof. The affine map has positive slope and the stated inverse $r = (t - a)/\mu$. Substituting $\gamma = \mu/a$ gives $\mu^2(1 + \gamma r)/(\mu + \gamma) = \mu t/(1 + a)$, proving (1), with $t = 1$ at r_* . The logarithm/exponential identities give (2). Integration by parts yields $\int_a^\infty te^{-t} dt = (1 + a)e^{-a}$, including $a = 0$. The change of variables proves both normalization assertions. Set $q = \gamma/(1 + \gamma r) > 0$. Differentiation gives $\Sigma_P' = q - \mu$ and $\Sigma_P'' = -q^2$. Solving these equations gives the recovery formulas, whose denominator is $(1 + \gamma r)^{-1} > 0$. At r_* one has $q = \mu$. $\square$

2 Calibrated analytic reconstruction

Theorem 2.1 (Curvature, Riccati and exact flow). *For twice differentiable $F : (0, \infty) \rightarrow \mathbb{R}$ with $F(1) = F'(1) = 0$, each of the following conditions identifies $F = H$:*

$$F''(t) = -t^{-2}, \quad F''(t) = -(F'(t) + 1)^2.$$

Equivalently, the marked exact flow

$$q(t+s) = \frac{q(t)}{1 + sq(t)}, \quad q(1) = 1,$$

for every pair $t, t+s > 0$ identifies $q(t) = 1/t$; with $F' = q - 1$ and $F(1) = 0$ it identifies $F = H$. More generally $F''(r) = -(r-b)^{-2}$ on $r > b$ identifies $F(r) = \log(r-b) + Ar + B$, and two independent affine data fix A, B .

Proof. Subtract the known logarithmic primitive and use the mean value theorem twice. For the Riccati equation put $q = F' + 1$ and $W(t) = (tq(t) - 1)e^{F(t)+t}$. Direct differentiation using $q' = -q^2$ gives $W' = 0$. The slope anchor gives $W(1) = 0$, so positivity of the exponential gives $tq(t) = 1$ everywhere. Integrating with the value anchor recovers F . For exact flow set $t = 1$, $s = v - 1$ directly. All forward assertions follow by differentiating H . Affine perturbations show why the two integration constants are not fixed by curvature alone. $\square$

Theorem 2.2 (Recentring transports its own regularity and anchors). *Let F be differentiable on $(0, \infty)$, with $F''(1) = -1$. Then*

$$F(av) - F(a) - aF'(a)(v-1) = F(v) \quad (a, v > 0)$$

holds if and only if $F = H$. No separate affine anchors or global second-differentiability hypothesis is needed.

Proof. Taking $v = 1$ gives $F(1) = 0$. Differentiate with respect to v at one to obtain $F'(1) = 0$. At arbitrary v differentiation gives $a(F'(av) - F'(a)) = F'(v)$. The difference quotient at $v = 1$ then gives existence of $F''(a)$ and $a^2F''(a) = F''(1) = -1$. Theorem 2.1 applies. Substitution verifies the converse. If the scale is not calibrated, every constant multiple of H satisfies the recentring equation. $\square$

Theorem 2.3 (Normalized group logarithm and cocycle). *On $D = (-1, \infty)$ define $x \star y = x + y + xy$. This is a commutative group with identity zero and inverse $-x/(1+x)$. A function $\ell : D \rightarrow \mathbb{R}$ satisfying*

$$\ell(x \star y) = \ell(x) + \ell(y), \quad \ell'(0) = 1$$

is uniquely $\ell(x) = \log(1+x)$, where differentiability is initially required only at zero. Equivalently the differentiable-at-zero cocycle clause

$$h(x \star y) = h(x) + h(y) - xy, \quad h'(0) = 0$$

identifies $h(x) = \log(1+x) - x = H(1+x)$.

Proof. The identity $1 + x \star y = (1 + x)(1 + y)$ proves closure, associativity, commutativity and the inverse formula. A homomorphism has $\ell(0) = 0$. For fixed x , write $x \star y = x + (1 + x)y$, divide its homomorphism identity by $(1 + x)y$, and let $y \rightarrow 0$. It gives $\ell'(x) = 1/(1 + x)$ everywhere. The anchor and the mean value theorem give the unique primitive. Adding the identity function to the cocycle turns it into exactly this homomorphism equation. Substitution proves the forward statements and the cocycle compatibility identity. $\square$

Theorem 2.4 (Typed completion equation). *Suppose h is differentiable on D , $h'(D) \subset D$ and $\ell = \text{id} + h$. Then*

$$\ell(h'(z)) = -\ell(z) \quad (z \in D)$$

holds if and only if, for some $K > 0$,

$$h(z) = K \log(1 + z) - z - \frac{K}{2} \log K.$$

Either anchor $h(0) = 0$ or $h'(0) = 0$ selects $K = 1$. Injectivity of ℓ and twice differentiability of h follow from the stated domain condition and equation.

Proof. Put $q = h'$. The domain condition gives $\ell' = 1 + q > 0$, so ℓ is strictly increasing and its inverse on its range is continuous. The equation $q = \ell^{-1} \circ (-\ell)$ makes q continuous. The inverse derivative theorem now makes q continuously differentiable. Applying the completion equation twice and using injectivity gives $q(q(z)) = z$. Differentiation of the first equation then gives $(1 + z)q'(z) = -(1 + q(z))$. Hence $(1 + z)(1 + q(z)) = K > 0$, and integration gives $h = K \log(1 + z) - z + C$. Substitution gives $2C = -K \log K$. Every displayed function has the requisite derivative range and satisfies the equation. Its two anchor values are $-(K/2) \log K$ and $K - 1$, each zero exactly when $K = 1$. $\square$

Proposition 2.5 (Automorphisms, integer series and their calibration). *Every continuous, measurable, monotone or differentiable automorphism of the fixed group $(D, \star)$ is*

$$F_c(x) = (1 + x)^c - 1, \quad c \in \mathbb{R} \setminus \{0\}.$$

The calibration $F'_c(0) = 1$ selects the identity. Without regularity, conjugates of arbitrary additive bijections of $\mathbb{R}$ give further automorphisms. Over a characteristic-zero field the formal automorphisms have the same formula, with nonzero field coefficient c . The integer series is $[n]_\star(x) = (1 + x)^n - 1$; it describes repeated group addition, and $[m]_\star \circ [n]_\star = [mn]_\star$.

Proof. Conjugate the automorphism by $\log(1 + x)$ and $e^u - 1$. It becomes an additive bijection. Each stated regularity condition makes an additive real function linear: integer and rational linearity follows algebraically, and continuity follows from measurability or local boundedness in the other cases. Rational approximation then gives $u \mapsto cu$, $c \neq 0$. A non-linear additive bijection constructed from a rational basis gives the unrestricted countermodel. In the formal case, conjugation by the normalized formal logarithm gives an additive

series. Comparing the mixed monomial $u^{n-1}v$ in its additivity identity annihilates every coefficient of degree $n > 1$, since n is invertible. The integer formula follows by the multiplicative coordinate $1 + x$. $\square$

Theorem 2.6 (Differentiable Bregman reconstruction). *For a differentiable $\Phi : (0, \infty) \rightarrow \mathbb{R}$, let $B_\Phi(t, s) = \Phi(t) - \Phi(s) - \Phi'(s)(t - s)$. Simultaneous scale invariance $B_\Phi(ct, cs) = B_\Phi(t, s)$ for all $c, t, s > 0$ holds if and only if*

$$\Phi(t) = AI(t) + B(t - 1) + C.$$

Thus $\Phi(1) = \Phi'(1) = 0$ and $\Phi'(2) - \Phi'(1) = 1/2$ identify $\Phi = I$ without separately assuming second differentiability or convexity. Alternatively, for an anchored differentiable Φ the exact symmetric slice

$$B_\Phi(t, 1) + B_\Phi(1, t) = t + t^{-1} - 2$$

identifies $\Phi = I$. In particular $B_I(t, s) = I(t/s)$.

Proof. Differentiate scale invariance only in its first argument, taking $s = 1$. For $g = \Phi' - \Phi'(1)$ one obtains $c(g(ct) - g(c)) = g(t)$. Swapping c, t and subtracting gives $g(c)(1 - 1/t) = g(t)(1 - 1/c)$. Fixing $c \neq 1$ yields $g(t) = A(1 - 1/t)$; integration gives the asserted affine family. The three calibrations fix its constants. For the symmetric slice, direct cancellation gives $(t-1)(\Phi'(t) - \Phi'(1)) = t + t^{-1} - 2$. For $t \neq 1$ divide by $t - 1$ to recover $\Phi' = 1 - 1/t$; the prescribed derivative at one supplies the remaining point. Integrate using $\Phi(1) = 0$. Direct substitution verifies the forward identities. Varying A, B, C separately proves the necessity of the three scalar calibrations within the scale-invariant family. $\square$

Theorem 2.7 (Exact Legendre target with either regularity condition). *Extend I by $+\infty$ off $(0, \infty)$. Its convex conjugate is*

$$I^*(\theta) = \begin{cases} -\log(1 - \theta), & \theta < 1, \\ +\infty, & \theta \geq 1. \end{cases}$$

If an extended-real function F has this exact full conjugate and is either lower semicontinuous on $(0, \infty)$ or convex, then $F = I$. It is unnecessary to assume both regularity conditions. Without either, the converse is false.

Proof. For $\theta < 1$, the maximizer of $\theta t - I(t)$ is $t = (1 - \theta)^{-1}$, giving the displayed value. The supremum diverges for $\theta \geq 1$. Fenchel's inequality for a candidate with this conjugate gives $F \geq I^{**} = I$, where the last equality also follows directly from the just computed maximizers. In particular $F = +\infty$ off the positive ray. For $t_0 > 0$, choose $\theta_0 = 1 - 1/t_0$. The function $(1 - \theta_0)t - 1 - \log t$ is coercive at both endpoints and has unique minimizer t_0 . A maximizing sequence in the definition of $F^*(\theta_0)$ must therefore converge to t_0 , with its F -values converging to $I(t_0)$. Lower semicontinuity gives the opposite inequality $F(t_0) \leq I(t_0)$. If F is instead convex, these sequences show its effective domain is dense in the positive ray; convexity and bracketing by two effective-domain points make it finite

everywhere there. A finite convex function on an open interval is continuous, so the same argument applies. Finally raise $I(2)$ by one and leave every other value unchanged. Nearby points approximate each supremum, so the conjugate is unchanged while the function is neither convex nor lower semicontinuous at two. $\square$

Theorem 2.8 (Self-concordance and the marked Hessian metric). *Let $\Phi \in C^3(0, \infty)$ with $\Phi'' > 0$, $\Phi(1) = \Phi'(1) = 0$ and $\Phi''(1) = 1$. The signed equality $\Phi''' = -2(\Phi'')^{3/2}$ identifies $\Phi = I$. The unsigned equality $|\Phi'''| = 2(\Phi'')^{3/2}$ also suffices on the entire positive ray. The corresponding Hessian metric is dt^2/t^2 , with distance $|\log(t/s)|$ and*

$$I(t/s) + I(s/t) = 4 \sinh^2 \left(\frac{|\log(t/s)|}{2} \right).$$

The metric in the marked affine coordinate together with the affine anchors reconstructs I ; its bare distance does not fix orientation. The self-concordance inequality alone does not identify I , and I is not a finite-parameter self-concordant barrier on this domain.

Proof. For the signed equality, $((\Phi'')^{-1/2})' = 1$, so calibration gives $\Phi'' = t^{-2}$ and the anchors give I . In the unsigned case the continuous nonzero third derivative has constant sign. The other sign would give $(\Phi'')^{-1/2} = 2 - t$, impossible on all $t > 0$. The coordinate $u = \log t$ makes the metric Euclidean, proving the distance formula; expanding $2 \cosh u - 2$ proves the symmetrization. Inversion $t \mapsto 1/t$ preserves the distance but reverses the directed ratio. The quadratic $\Phi(t) = (t - 1)^2/2$ satisfies all three anchors, has positive second derivative, and satisfies the usual self-concordance inequality since its third derivative is zero, yet differs from I . Moreover, $(I')^2/I'' = (t - 1)^2$ is unbounded, refuting a finite barrier parameter. The affine-equivalent function $-\log t$ does have barrier parameter one. $\square$

Theorem 2.9 (Calibrated projective and Schwarzian data). *An injective real-valued function f on $(0, \infty)$ preserving all cross-ratios and satisfying $f(1) = 0$, $f(2) = -1/2$, $f(3) = -2/3$ equals $1/t - 1$. Consequently $f = F'$ and $F(1) = 0$ identify $F = H$. Alternatively, if $f \in C^3$, $f' \neq 0$,*

$$\frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2 = 0, \quad f(1) = 0, \quad f'(1) = -1, \quad f''(1) = 2,$$

then the same conclusion holds. Projectivity or zero Schwarzian without calibration does not select this member.

Proof. With cross-ratio convention $(x_1 - x_3)(x_2 - x_4)/((x_1 - x_4)(x_2 - x_3))$, fix the first three arguments at 1, 2, 3. The cross-ratio is injective fractional-linear in the fourth argument. Comparing f with $1/t - 1$ identifies every other point; the marks identify the three excluded ones. For the Schwarzian version put $w = f''/f'$. Its equation is $w' = w^2/2$ with $w(1) = -2$, whence $w = -2/t$

by ODE uniqueness. Integrate twice with the two remaining marks to obtain $f' = -t^{-2}$ and $f = t^{-1} - 1$. Integrating the anchored primitive yields H . Without the marks other fractional-linear functions remain. $\square$

Proposition 2.10 (Derivative, discrete and local-data boundaries). *The placed derivatives satisfy*

$$\Sigma_P^{(n)}(r) = (-1)^{n-1}(n-1)! \left(\frac{\gamma}{1+\gamma r} \right)^n, \quad n \geq 2.$$

All orders beyond two add no information once curvature is known. A lone prescribed derivative of order n identifies a function only modulo a polynomial of degree at most $n-1$. For $\rho(r) = C(1+\gamma r)e^{-\mu r}$, $C > 0$, its derivative of order $n \geq 1$ has the single zero $r_n = n/\mu - 1/\gamma$. Within this shape family, two consecutive indexed zeros determine μ, γ , and mass determines C . The discrete curvature is

$$\Sigma_P(r+1) - 2\Sigma_P(r) + \Sigma_P(r-1) = \log \left(1 - \frac{1}{(r-b)^2} \right), \quad r-1 > b.$$

It determines lattice values from two starting values, not an arbitrary real interpolation. A complete germ identifies a global function in a specified connected real-analytic class, but not in a general smooth class.

Proof. Differentiate the logarithm inductively. Repeated integration proves the polynomial freedom and Theorem 2.1 proves the second-derivative sufficiency. Leibniz's rule gives

$$\rho^{(n)} = C(-\mu)^{n-1}e^{-\mu r} [n\gamma - \mu(1+\gamma r)],$$

proving the exact zero. Differences of consecutive zeros recover $1/\mu$ and then the intercept $-1/\gamma$. Products of the three logarithmic arguments give the discrete formula; recurrence gives the lattice-value statement. A nonzero multiple of $\sin(2\pi r)$ preserves all integer samples and changes interpolation. A sufficiently small compactly supported smooth bump away from an observed point preserves its entire jet. Two disjoint bumps with cancelling integrals preserve normalization of a positive density, and can be supported away from the germ, the mode and endpoints. Analytic continuation instead uses the identity theorem on the common connected domain. $\square$

Theorem 2.11 (Exact all-order derivative-zero sets do not identify Σ). *For every real $k > 1$ the function*

$$f_k(t) = (k-1)k^k \frac{t}{(t+k)^{k+1}}, \quad t > 0,$$

is a positive real-analytic probability density, vanishing at both endpoints, and for every integer $n \geq 1$,

$$f_k^{(n)}(t) = (-1)^n (k-1)k^k (k)_n \frac{t-n}{(t+k)^{k+n+1}}.$$

Here $(k)_n = k(k+1)\cdots(k+n-1)$. Thus every derivative has exactly the same single simple positive zero n as $p^{(n)}$, but $f_k \neq p$. In particular $4t/(t+2)^3$ shows that these derivative-zero data do not characterize p , even among normalized real-analytic positive densities.

Proof. Substitution $t = ks$ and subtraction of two elementary power integrals give

$$\int_0^\infty \frac{t dt}{(t+k)^{k+1}} = k^{1-k} \int_0^\infty \frac{s ds}{(1+s)^{k+1}} = \frac{k^{-k}}{k-1}.$$

This proves normalization; positivity and real analyticity follow on the positive ray. The formula is true at order zero if $(k)_0 = 1$. Its induction step is the exact identity

$$\frac{d}{dt} \frac{t-n}{(t+k)^{k+n+1}} = -(k+n) \frac{t-(n+1)}{(t+k)^{k+n+2}}.$$

Every factor except $t-n$ is nonzero on $t > 0$, proving exactness and simplicity of the zero set, including the derivative sign pattern. The algebraic tail separates f_k from te^{-t} . The counterexample is not entire and is not globally log-concave: $(\log f_k)'' = -t^{-2} + (k+1)/(t+k)^2 > 0$ when $t > 1 + \sqrt{k+1}$. These additional conditions are not premises of the refuted characterization. Under $t = \mu r + a$, change variables and normalize on the selected half-line; multiplication by a positive constant leaves every placed derivative zero $(n-a)/\mu$ unchanged. $\square$

3 Probability laws and identifying transforms

Throughout this section we use the intrinsic decomposition of Σ in its marked positive coordinate:

$$H(t) = \log t - t + 1, \quad I(t) = t - 1 - \log t, \quad p(t) = te^{-t}.$$

The symbol $\bar{F}$ denotes a survival function. All Gamma parameters are shape and *rate*, in that order. Equality of probability laws identifies densities almost everywhere; a pointwise density assertion uses their specified continuous representatives. Placement parameters do not occur in these intrinsic statements.

Proposition 3.1 (Uniqueness principles for the transforms used below). *Finite Borel measures on $\mathbb{R}$ are determined by their characteristic functions. Finite measures on $[0, \infty)$ are determined by their Laplace transforms on $[0, \infty)$, and finite measures on $[0, 1]$ by all their nonnegative integer moments. Two probability laws with moment generating functions finite on an open interval about zero and equal there are equal.*

Proof. For the compact-interval assertion, equality on monomials gives equality on polynomials, then on continuous functions by uniform polynomial approximation, and finally on Borel sets by uniqueness of finite regular measures. Pushing a measure on $[0, \infty)$ forward by $x \mapsto e^{-x}$ reduces Laplace uniqueness to this assertion: integer Laplace values give all moments of the pushforward on $[0, 1]$.

For characteristic functions, subtract the two finite measures. Convolution with a centered Gaussian of variance $\varepsilon > 0$ has density whose Fourier transform is the vanishing characteristic function times $e^{-\varepsilon\xi^2/2}$. Fourier inversion, justified by the integrable Gaussian factor and Fubini's theorem, makes this convolution zero. Gaussian approximate identities converge against every continuous function of compact support, so the original signed measure is zero. Finally an exponential moment gives a holomorphic bilateral transform on a vertical strip about the imaginary axis. The identity theorem extends equality near zero to that strip; characteristic-function uniqueness applies on the imaginary axis. $\square$

Theorem 3.2 (Gamma law and complete transform characterizations). *Let $\mu_\Gamma(dt) = \mathbf{1}_{(0,\infty)}(t)p(t)dt$. It is the probability law $\text{Gamma}(2, 1)$, with topological support $[0, \infty)$ and no atom at zero. Its transforms are*

$$\int e^{-zt} \mu_\Gamma(dt) = (1+z)^{-2}, \quad \text{Re } z > -1, \quad (3)$$

$$\int e^{i\xi t} \mu_\Gamma(dt) = (1-i\xi)^{-2}, \quad \xi \in \mathbb{R}, \quad (4)$$

$$\int t^{z-1} \mu_\Gamma(dt) = \Gamma(z+1), \quad \text{Re } z > -1, \quad (5)$$

$$\int t^n \mu_\Gamma(dt) = (n+1)!, \quad n \in \mathbb{N}_0. \quad (6)$$

Each of the following complete observations identifies its candidate as μ_Γ :

1. a finite positive Borel measure on $\mathbb{R}$ with the moments (6), including $n = 0$;
2. a probability on $[0, \infty)$ with Laplace transform $(1+\lambda)^{-2}$ for every $\lambda \geq 0$;
3. a probability on $\mathbb{R}$ with characteristic function $(1-i\xi)^{-2}$ for every real ξ ;
4. a probability on $(0, \infty)$ with $\mathbb{E}T^{i\xi} = \Gamma(2+i\xi)$ for every real ξ .

In the first and third clauses, positive support is a conclusion. Moreover

$$\mathbb{E}T = 2, \quad \text{Var}(T) = 2, \quad \mathbb{E} \log T = 1 - \gamma_E, \quad \kappa_n(T) = 2(n-1)! \quad (n \geq 1).$$

Proof. Euler's integral and integration by parts give $\int_0^\infty t^a e^{-bt} dt = \Gamma(a+1)b^{-a-1}$ for $a > -1$, $b > 0$, with its holomorphic extension on the stated half-planes. Taking $a = 1$ proves normalization and the first two transforms; taking $a = z$ proves the Mellin formula and the moments. Differentiating the Gamma integral at $a = 1$ is justified by an integrable logarithmic majorant near zero and exponential decay at infinity. It gives $\mathbb{E} \log T = \Gamma'(2) = 1 - \gamma_E$. The identity $\log \mathbb{E}e^{uT} = -2 \log(1-u)$ yields the cumulants.

For the first converse the zeroth moment gives mass one. For $0 < c < 1$, positivity and monotone convergence give

$$\int \cosh(ct) \mu(dt) = \sum_{n=0}^{\infty} (2n+1)c^{2n} = \frac{1+c^2}{(1-c^2)^2} < \infty.$$

Since $e^{c|t|} \leq 2 \cosh(ct)$, the candidate has a two-sided exponential moment. Dominated power-series expansion for $|u| < 1$ therefore gives

$$\int e^{ut} \mu(dt) = \sum_{n=0}^{\infty} \frac{u^n (n+1)!}{n!} = (1-u)^{-2}.$$

Proposition 3.1 identifies the measure on the whole real line. Its support follows from the strictly positive density on $(0, \infty)$ and zero density on $(-\infty, 0]$. The second and third converses are the same uniqueness proposition. In the fourth, the supplied function is the characteristic function of $\log T$; identify that law and push it forward by the exponential. $\square$

Proposition 3.3 (Exponential samples also derive support). *If a finite positive Borel measure ν on $\mathbb{R}$ satisfies*

$$\int e^{-ns} \nu(ds) = (n+1)^{-2} \quad (n \in \mathbb{N}_0),$$

then $\nu(ds) = \mathbf{1}_{(0,\infty)}(s) s e^{-s} ds$.

Proof. The sample at zero gives mass one. If $\nu((-\infty, -\varepsilon]) = m > 0$, then the n th sample is at least $m e^{n\varepsilon}$, a contradiction as $n \rightarrow \infty$. Taking a countable union excludes the entire negative ray. Push forward by $y = e^{-s}$ to $[0, 1]$. Its moments are $(n+1)^{-2}$, which are those of $(-\log y) dy$ on $(0, 1)$. Compact moment uniqueness and the inverse change of variables give the stated law, including absence of an atom at zero. $\square$

Proposition 3.4 (Boundaries of moment identification). *Removing positivity, removing the zeroth moment, or retaining only finitely many ordinary moments invalidates the corresponding unrestricted moment converse in Theorem 3.2.*

Proof. Without the zeroth equation, every $\mu_{\Gamma} + a\delta_0$, $a > 0$, has the specified moments of positive orders. To remove positivity, take a nonzero real even smooth Fourier bump supported in $[-2, -1] \cup [1, 2]$ and let h be its inverse Fourier transform. The real Schwartz function h has every polynomial moment zero, because those moments are constant multiples of derivatives of its Fourier transform at zero. Thus the distinct finite signed measures $\mu_{\Gamma} + \varepsilon h(t) dt$ have all the same moments.

For any fixed m , choose $m+2$ nonzero smooth bumps with disjoint compact supports in $(1, 2)$. The $m+1$ linear equations making a linear combination h orthogonal to $1, t, \dots, t^m$ have a nonzero solution. Disjoint supports ensure $h \neq 0$. On this compact set p has a positive minimum, so $p + \varepsilon h$ is a positive probability density for all sufficiently small $|\varepsilon|$. It is distinct from p and retains every requested moment. $\square$

Theorem 3.5 (Marked Poisson recurrence, cumulants and divergences). *A probability sequence $(\pi_n)_{n \geq 0}$ satisfies*

$$(n+1)\pi_{n+1} = \pi_n \quad (n \geq 0)$$

if and only if $\pi_n = e^{-1}/n!$. For N with this law,

$$K(u) = \log \mathbb{E} e^{uN} = e^u - 1, \quad (7)$$

$$\Lambda(u) = \log \mathbb{E} e^{u(N-1)} = e^u - 1 - u = I(e^u). \quad (8)$$

The complete uncentered CGF near zero identifies N as $\text{Pois}(1)$. The complete centered expression identifies the law of $N - 1$ when the shift is marked as one. As a scalar function it reconstructs $I(t) = \Lambda(\log t)$.

With D denoting relative entropy, for $a, b > 0$ one has

$$D(\text{Pois}(a) \parallel \text{Pois}(b)) = aI(b/a), \quad (9)$$

$$D(\text{Exp}(a) \parallel \text{Exp}(b)) = I(b/a), \quad (10)$$

$$2D(\mathcal{N}(0, a) \parallel \mathcal{N}(0, b)) = I(a/b), \quad (11)$$

where a, b in the last line are variances. Consequently each full marked slice obtained by $(a, b) = (1, t), (1, t), (t, 1)$, respectively, reconstructs I . With $\Psi(t) = t \log t - t + 1$,

$$D(\text{Pois}(a) \parallel \text{Pois}(b)) = \Psi(a) - \Psi(b) - \Psi'(b)(a - b).$$

The Legendre transform of the centered CGF is

$$\Lambda^*(z) = \begin{cases} (1+z) \log(1+z) - z, & z > -1, \\ 1, & z = -1, \\ +\infty, & z < -1. \end{cases}$$

Proof. The recurrence gives $\pi_n = \pi_0/n!$ by induction; summing fixes $\pi_0 = e^{-1}$. Conversely these coefficients satisfy the recurrence. Summing the exponential series gives $\mathbb{E} e^{uN} = e^{-1} \exp(e^u)$ and both CGFs. Their converses follow from Proposition 3.1 applied to the actual variable whose CGF is supplied.

For Poisson laws the log likelihood ratio at n is $b - a + n \log(a/b)$; its expectation is $b - a + a \log(a/b)$. For exponential laws the log ratio is $\log(a/b) + (b - a)t$, whose expectation under rate a is $\log(a/b) + (b - a)/a$. For centered Gaussians the log ratio is $\frac{1}{2} \log(b/a) + \frac{1}{2} t^2 (1/b - 1/a)$, whose expectation uses $\mathbb{E} t^2 = a$. Substitution proves the three formulas and the Bregman identity.

For $z > -1$, maximizing $uz - \Lambda(u)$ gives $e^u = 1 + z$ and the displayed value. At $z = -1$ the objective is $1 - e^u$ with supremum one as $u \rightarrow -\infty$. For $z < -1$ it tends to infinity in that limit. Exponential tilting by uN gives $\text{Pois}(e^u)$, whose mean, variance, and Fisher information in the u coordinate are all e^u , as follows by differentiating K and by the score $N - e^u$. $\square$

Proposition 3.6 (The centering mark and KL orientation are essential). *An unmarked centered CGF does not determine the original location, even for a law on $\mathbb{N}_0$. Reversing the displayed divergence slices changes their scalar potential.*

Proof. Let $Y \sim \text{Pois}(1)$ and $N_k = Y + k$ for an integer $k \geq 1$. Then $N_k - \mathbb{E} N_k = Y - 1$, with centered CGF $e^u - 1 - u$, although N_k is not $\text{Pois}(1)$. The reversed

Poisson slice is $t \log t - t + 1$; the reversed exponential slice is $\log t + t^{-1} - 1 = I(1/t)$, and twice the reversed Gaussian slice has this same value. These follow directly from (9)–(11) and differ from $I(t)$ in general. $\square$

Theorem 3.7 (Two marked max laws characterize the Gumbel CDF). *Let F be a CDF on $\mathbb{R}$. The identities*

$$F(x + \log 2)^2 = F(x), \quad F(x + \log 3)^3 = F(x) \quad (x \in \mathbb{R}), \quad F(0) = e^{-1}$$

hold if and only if

$$F(x) = \exp(-e^{-x}).$$

No density, smoothness, strict monotonicity, or full-support hypothesis is required. This CDF has density $g(x) = e^{-x-e^{-x}} = p(e^{-x})$, and it satisfies the analogous max identity for every positive integer n .

Proof. Iteration in both directions gives $F(k \log 2) = \exp(-2^{-k})$ for every integer k . Monotone bounds by adjacent such values yield $0 < F(x) < 1$ at every finite x . Hence $G = -\log F$ is finite, positive, and nonincreasing. The two equations imply

$$G(m \log 2 + n \log 3) = e^{-(m \log 2 + n \log 3)} \quad (m, n \in \mathbb{Z}).$$

The subgroup generated by $\log 2$ and $\log 3$ is dense in $\mathbb{R}$: otherwise their ratio would be rational, contradicting unique factorization in the equality $2^r = 3^s$ for positive integers r, s . Approximating an arbitrary x from both sides by subgroup points and using monotonicity proves $G(x) = e^{-x}$. Substitution proves the converse, the density formula, and the max identities. For independent observations the CDF of their maximum is F^n , so these are precisely the stated shifted max laws. $\square$

Proposition 3.8 (Haar weighting and Gumbel calibration boundaries). *For the marked map $\tau(x) = e^{-x}$, the Gumbel probability satisfies*

$$\tau_*(dF) = e^{-t} dt = p(t) \frac{dt}{t}, \quad \int_0^\infty p(t) \frac{dt}{t} = 1.$$

The measure dt/t is multiplicative Haar measure and has infinite mass. Thus the Haar-weighted probability, and its complete density in the marked logarithmic coordinate, each reconstruct p . Removing the anchor in Theorem 3.7, or keeping only its base-two equation, destroys uniqueness.

Proof. The positive change-of-variables Jacobian is $|dx/dt| = 1/t$. Since $g(-\log t) = p(t)$, it gives the pushforward formula; conversely $p(t) = g(-\log t)$ recovers the density. Invariance of dt/t under $t \mapsto at$ is immediate by substitution, and its mass diverges at both endpoints.

Without the anchor, $F_c(x) = \exp(-ce^{-x})$, $c > 0$, satisfies both max equations. For only the base-two equation, put $a = \log 2$ and $h(x) = 1 + \varepsilon \sin(2\pi x/a)$. For nonzero ε sufficiently small, $h > 0$ and $h' < h$. Then $\exp(-e^{-x}h(x))$ is a strictly increasing CDF with the correct endpoint limits and anchor; periodicity gives the base-two equation. It is not the Gumbel CDF. The original Gamma probability $p(t) dt$ and the Haar-weighted probability $p(t) dt/t$ are different measures. $\square$

Proposition 3.9 (Canonical deficit law and coordinate involution). *For $T \sim \mu_\Gamma$ let $V = I(T)$. For $v \geq 0$ let $a(v) \leq 1 \leq b(v)$ be the roots of $I(t) = v$; explicitly*

$$a(v) = -W_0(-e^{-1-v}), \quad b(v) = -W_{-1}(-e^{-1-v}),$$

where W_0 and W_{-1} are the real inverse branches of we^w . Writing $\bar{F}_\Gamma(t) = (1+t)e^{-t}$, the canonical deficit CDF is zero for $v < 0$ and is

$$F_I(v) = \bar{F}_\Gamma(a(v)) - \bar{F}_\Gamma(b(v)), \quad v \geq 0. \quad (12)$$

It has no atom at zero and, for $v > 0$, has density

$$f_I(v) = e^{-1-v} \left(\frac{a(v)}{1-a(v)} + \frac{b(v)}{b(v)-1} \right), \quad f_I(v) \sim \frac{\sqrt{2}}{e\sqrt{v}} \quad (v \downarrow 0). \quad (13)$$

The canonical coordinate involution is

$$j_I(t) = \begin{cases} -W_{-1}(-te^{-t}), & 0 < t < 1, \\ 1, & t = 1, \\ -W_0(-te^{-t}), & t > 1. \end{cases} \quad (14)$$

It is a smooth strictly decreasing involution, with $j'_I(1) = -1$, unique fixed point one, and $I \circ j_I = I$, $p \circ j_I = p$.

Proof. The derivative $I'(t) = (t-1)/t$ and the infinite endpoint limits give precisely two roots at every positive level. Rearranging $I(t) = v$ gives $(-t)e^{-t} = -e^{-1-v}$ and proves the Lambert formulas. The sublevel set is $[a(v), b(v)]$, proving the CDF by integration. Implicit differentiation gives $a' = a/(a-1)$ and $b' = b/(b-1)$, while $p(a) = p(b) = e^{-1-v}$; differentiating the CDF gives its density. The Taylor expansion $I(1+h) = h^2/2 + O(h^3)$ gives $a(v) = 1 - \sqrt{2v} + O(v)$ and $b(v) = 1 + \sqrt{2v} + O(v)$, yielding the stated asymptotic. The zero level is the singleton $\{1\}$, which has Gamma mass zero.

Equal-level pairing immediately gives the involution identities. To verify smoothness at the joining point, write $I(t) = (t-1)^2 r(t)$ near one, where r is smooth, positive, and $r(1) = 1/2$. The function

$$q_I(t) = \text{sgn}(t-1)\sqrt{2I(t)}$$

therefore extends smoothly with derivative one at one. Away from one its derivative is strictly positive by the sign of I' . The endpoint limits make it a smooth increasing bijection onto $\mathbb{R}$, and $j_I = q_I^{-1} \circ (-\text{id}) \circ q_I$. This proves all the remaining assertions, including $j'_I(1) = -1$. $\square$

Proposition 3.10 (Deficit transform, cumulants, and determinacy). *The maximal real moment-generating domain of V is $s < 1$, and*

$$M_V(s) = e^{-s} \frac{\Gamma(2-s)}{(1-s)^{2-s}}, \quad (15)$$

$$K_V(s) = \gamma_{Es} + \sum_{n=2}^{\infty} \frac{1}{n} \left(\zeta(n) - \frac{1}{n-1} \right) s^n \quad (|s| < 1). \quad (16)$$

Consequently

$$\kappa_1(V) = \gamma_E, \quad \kappa_n(V) = (n-1)! \left(\zeta(n) - \frac{1}{n-1} \right) \quad (n \geq 2). \quad (17)$$

In particular $\text{Var } V = \pi^2/6 - 1$, $\kappa_3(V) = 2\zeta(3) - 1$, and $\kappa_4(V) = \pi^4/15 - 2$. Every nonnegative probability law with all finite moments and these algebraic cumulants equals the law of V ; no separate exponential moment assumption on that candidate is necessary. For $a > 0$ and $s < 1$, the conditional transform is

$$\mathbb{E}[e^{sI(T)} \mid T \geq a] = \frac{e^{-s}\Gamma(2-s, (1-s)a)}{(1+a)e^{-a}(1-s)^{2-s}}, \quad (18)$$

where $\Gamma(q, x) = \int_x^\infty u^{q-1} e^{-u} du$.

Proof. The identity $p = e^{-1-I}$ gives

$$M_V(s) = e^{-s} \int_0^\infty t^{1-s} e^{-(1-s)t} dt.$$

For $s < 1$ the Gamma substitution gives (15). At $s = 1$ its integrand is the positive constant e^{-1} , and for $s > 1$ the tail diverges. Euler's Gamma product, expanded absolutely for $|s| < 1$, gives

$$\log \Gamma(1-s) = \gamma_E s + \sum_{k=1}^\infty \{-\log(1-s/k) - s/k\} = \gamma_E s + \sum_{n=2}^\infty \zeta(n) s^n / n.$$

Using $\Gamma(2-s) = (1-s)\Gamma(1-s)$ and the logarithm series in $\log M_V$ proves (16), and differentiation gives the cumulants. The classical evaluations $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$ give the displayed values.

Algebraic cumulants and moments determine each other recursively:

$$m_0 = 1, \quad m_{n+1} = \sum_{j=0}^n \binom{n}{j} \kappa_{j+1} m_{n-j}.$$

This is coefficient comparison in the formal identity $M' = K'M$. Thus a candidate has the same moments as V . For any $0 < s < 1$, monotone convergence of its nonnegative exponential series gives $\mathbb{E}e^{sX} = \sum m_n s^n / n! = M_V(s) < \infty$. Nonnegativity also gives finiteness for negative s ; uniqueness from Proposition 3.1 finishes the converse. Restricting the same integral to $[a, \infty)$ and dividing by $\mathbb{P}(T \geq a) = (1+a)e^{-a}$ proves the conditional formula. $\square$

Theorem 3.11 (The linked deficit law and involution identify the potential). *Let $J : (0, \infty) \rightarrow [0, \infty)$ be continuous, satisfy $J(1) = 0$, be strictly decreasing on $(0, 1]$ and strictly increasing on $[1, \infty)$, and tend to infinity at both endpoints. Suppose $q_J(t) = e^{-1-J(t)}$ has integral one. Let F_J be the law of $J(T_J)$ under this same density q_J , and let j_J exchange its two equal-level roots in the same t coordinate and fix one. If*

$$F_J = F_I \quad \text{and} \quad j_J = j_I,$$

with the target values specified in (12) and (14), then $J = I$ and $q_J = p$. Conversely I supplies these data. The full list (17) may replace $F_J = F_I$. More generally the complete pair (F_J, j_J) uniquely determines every potential in this candidate class.

Proof. Let $a_J(v) \leq 1 \leq b_J(v)$ be the inverse branches, including $a_J(0) = b_J(0) = 1$, and put $w_J(v) = b_J(v) - a_J(v)$. The pushforward of Lebesgue measure under J has locally finite Stieltjes distribution w_J : the sublevel interval has exactly that length. Weighting by e^{-1-J} therefore gives, as measures,

$$dF_J(v) = e^{-1-v} dw_J(v), \quad w_J(v) = e \int_{[0,v]} e^s dF_J(s).$$

Both widths vanish at zero, so there is no undetermined integration constant. The map $D_J(t) = t - j_J(t)$ is a continuous strictly increasing bijection from $[1, \infty)$ to $[0, \infty)$: j_J is decreasing, fixes one, and tends to zero as $t \rightarrow \infty$. Consequently

$$b_J(v) = D_J^{-1}(w_J(v)), \quad a_J(v) = j_J(b_J(v)).$$

The observed pair recovers both inverse branches and hence J pointwise. This proof uses Stieltjes measures and does not require a derivative of J or a density of F_J . The cumulant alternative follows from Proposition 3.10, since $J(T_J) \geq 0$. $\square$

Proposition 3.12 (Neither linked observation alone suffices). *Within the normalized candidate class of Theorem 3.11, there are noncanonical potentials with exactly F_I , and others with exactly j_I . Thus both canonical target observations and their common density–potential linkage carry identifying information. The law of $I(T)$ alone does not identify an arbitrary source law for T , even when the potential I itself is fixed.*

Proof. Start with the canonical inverse branches a, b . Choose a nonzero smooth function h supported on a compact level interval inside $(0, \infty)$ and set $a_\varepsilon = a + \varepsilon h$, $b_\varepsilon = b + \varepsilon h$. On that compact interval the strictly negative derivative of a and strictly positive derivative of b are bounded away from zero, and the branches are separated from zero and one. Small $|\varepsilon|$ therefore preserves the branch ranges and monotonicities. Their inverses define an admissible $J_\varepsilon \neq I$. The width is unchanged; the Stieltjes formula in Theorem 3.11 preserves both the entire deficit law and its total probability mass. Joint uniqueness shows that its involution differs.

Conversely choose nonzero nonnegative smooth level bumps f, g with disjoint compact supports in $(0, \infty)$, and set $h_{\varepsilon, \delta}(v) = v + \varepsilon f(v) + \delta g(v)$. Small coefficients make h a strictly increasing bijection fixing zero, with $h'(0) = 1$. The potential $h \circ I$ has exactly j_I . Its mass functional

$$N(\varepsilon, \delta) = \int_0^\infty e^{-1-h_{\varepsilon, \delta}(I(t))} dt$$

is smooth near zero, since the perturbations have compact level support, and $\partial_\delta N(0, 0) = -\int p(t)g(I(t)) dt < 0$. The implicit function theorem gives $\delta(\varepsilon)$

with $N(\varepsilon, \delta(\varepsilon)) = 1$. For nonzero small ε , disjoint supports ensure that this potential differs from I , and joint uniqueness makes its deficit law different.

Finally draw V with law F_I and set $T_+ = b(V)$. Then $I(T_+) = V$ exactly, but $T_+ \geq 1$ almost surely, so its law is not μ_Γ . This explicitly exhibits the lost allocation between the two branches if the source law is not linked to $e^{-1-I}dt$. Merely possessing some linked pair, without prescribing its two canonical values, plainly holds for all the potentials just built. $\square$

Theorem 3.13 (Survival, hazard, logistic equation, and causal Green kernel). *The Gamma survival and hazard are*

$$\overline{F}_\Gamma(x) = (1+x)e^{-x} = \frac{p(1+x)}{p(1)} = e^{-I(1+x)}, \quad h_\Gamma(x) = \frac{x}{1+x} \quad (x \geq 0). \quad (19)$$

Each of the following identifying clauses recovers p :

1. *an absolutely continuous probability on $[0, \infty)$ with positive survival at finite arguments, survival value one at zero, and actual hazard $x/(1+x)$ almost everywhere;*
2. *that same survival framework, with its marked hazard $v(u) = h(e^u)$ differentiable on $\mathbb{R}$ and satisfying $v' = v(1-v)$, $v(0) = 1/2$;*
3. *a C^2 function f on $(0, \infty)$ extending continuously to zero, with $f'' + 2f' + f = 0$, $f(0) = 0$, and $\int_0^\infty f(t) dt = 1$;*
4. *a distribution g on $\mathbb{R}$ supported in $[0, \infty)$ with $(D+1)^2 g = \delta_0$.*

*In the last clause the result is $g(t) = \mathbf{1}_{[0, \infty)}(t)te^{-t}$. If $k(t) = \mathbf{1}_{[0, \infty)}(t)e^{-t}$, then $g = k * k$. Thus μ_Γ is also the law of a sum of two independent unit-rate exponential variables and the second arrival time of a unit-rate Poisson process.*

Proof. Integration by parts gives the survival formula, and its negative derivative divided by the survival gives the hazard. For the first converse, local absolute continuity and positivity imply

$$(\log \overline{F})' = -\frac{x}{1+x} \quad \text{almost everywhere.}$$

Integration from zero gives $\log \overline{F}(x) = -x + \log(1+x)$; differentiating recovers the density p almost everywhere. The function $v(u) = (1 + e^{-u})^{-1}$ solves the second clause. Uniqueness for the scalar differential equation, whose right-hand side is locally Lipschitz, gives this solution on all of $\mathbb{R}$; the substitution $x = e^u$ returns the first clause.

For the third clause, $(e^t f)'' = 0$, so $f(t) = (A + Bt)e^{-t}$. The endpoint gives $A = 0$ and integration gives $B = 1$. For the distributional clause, multiplication by e^t conjugates $(D+1)^2$ to D^2 , while $e^t \delta_0 = \delta_0$. The function $t_+ = \max(t, 0)$ has distributional second derivative δ_0 . The difference $e^t g - t_+$ has second derivative zero, hence is an affine distribution; its support in $[0, \infty)$ forces both affine

coefficients to vanish. This proves existence and uniqueness. Direct convolution gives

$$(k * k)(t) = \mathbf{1}_{[0, \infty)}(t) \int_0^t e^{-s} e^{-(t-s)} ds = \mathbf{1}_{[0, \infty)}(t) t e^{-t}.$$

The convolution is the independent-sum density. For a unit-rate Poisson process the event that its second arrival exceeds x is the event of at most one arrival by x , with probability $e^{-x}(1+x)$. This gives the final construction. $\square$

Proposition 3.14 (Precise survival boundaries and Gamma shift uniqueness). *Local absolute continuity and the survival anchor in Theorem 3.13 cannot be dropped from its almost-everywhere hazard formulation. For the logarithmic random variable $Y = \log T$, the hazard with respect to its own coordinate is*

$$h_Y(u) = e^u h_\Gamma(e^u) = \frac{e^{2u}}{1 + e^u},$$

which differs from the composition $v(u)$ in that theorem. For a Gamma density of shape $\alpha > 0$ and rate $\beta > 0$, the self-survival shift identity

$$\overline{F}_{\alpha, \beta}(x) = \frac{f_{\alpha, \beta}(x + c)}{f_{\alpha, \beta}(c)} \quad (x \geq 0), \quad c > 0, \quad (20)$$

holds exactly when $\alpha = 1$ with arbitrary c , or when $\alpha = 2$ and $c = 1/\beta$.

Proof. The mixtures $r\delta_0 + (1-r)\mu_\Gamma$, $0 < r < 1$, retain the hazard at positive arguments and change the survival anchor. For the regularity boundary, let C be a nonconstant continuous nondecreasing singular function on $[0, \infty)$, zero on $[0, 1]$, constant on $[2, \infty)$, and with derivative zero almost everywhere. For instance, translate a Cantor distribution function to $[1, 2]$. Then

$$\overline{F}(x) = (1+x)e^{-x}e^{-C(x)}$$

is a continuous decreasing survival with value one at zero and limit zero, and satisfies the same logarithmic derivative equation almost everywhere. Its singular component makes it different from $\overline{F}_\Gamma$. This example concerns a bare derivative equation without absolute continuity. For $Y = \log T$, the density is $e^u p(e^u)$ and the survival is $\overline{F}_\Gamma(e^u)$, which gives the asserted Jacobian factor.

For (20), the density is $f_{\alpha, \beta}(x) = \beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$. If $0 < \alpha < 1$, the right derivative of the survival at zero is $-\infty$, while the shifted ratio has finite derivative, so equality is impossible. If $\alpha = 1$, the ratio and survival are both $e^{-\beta x}$ for every c . For $\alpha > 1$, the survival derivative at zero is zero; equality forces $(\alpha - 1)/c - \beta = 0$. For that c , the logarithmic derivative of the ratio is $-\beta x/(x+c)$, so its derivative is $-(\beta/c)x(1+o(1))$ near zero. The survival derivative is $-\beta^\alpha x^{\alpha-1}(1+o(1))/\Gamma(\alpha)$. Their equality forces $\alpha - 1 = 1$, then $c = 1/\beta$. The Gamma shape-two survival is $(1 + \beta x)e^{-\beta x}$, which verifies the surviving case. In particular this shift property without a specified comparison class is not an unrestricted characterization of p . $\square$

Proposition 3.15 (Probability Stieltjes transform and its inverse). *For $z > 0$,*

$$G_\Gamma(z) = \int_0^\infty \frac{p(t)}{t+z} dt = 1 - ze^z E_1(z), \quad E_1(z) = \int_z^\infty \frac{e^{-u}}{u} du. \quad (21)$$

The complete transform, or its values on any nonempty open interval in $(0, \infty)$, identifies a candidate finite positive measure on $[0, \infty)$ as μ_Γ . The Laplace transform $(1+\lambda)^{-2}$ is completely monotone but is not a nonzero Stieltjes function. Neither p nor $\bar{F}_\Gamma$ is completely monotone.

Proof. Split $t/(t+z) = 1 - z/(t+z)$ and substitute $u = t+z$ in the second integral; this proves (21). For a finite measure μ , Tonelli's theorem gives

$$G_\mu(z) = \int_0^\infty e^{-zv} L_\mu(v) dv, \quad L_\mu(v) = \int e^{-vt} \mu(dt).$$

The Stieltjes transform is holomorphic for $\operatorname{Re} z > 0$, by local dominated differentiation. Equality on an open positive interval thus extends to all positive z . For two such measures let $h = L_\mu - L_\nu$, a bounded continuous function. The finite signed measure $e^{-v} h(v) dv$ has zero Laplace transform by the displayed identity evaluated at $z+1$. Proposition 3.1 makes it zero; continuity gives $h = 0$, and a second application of Laplace uniqueness gives $\mu = \nu$.

Complete monotonicity of $(1+\lambda)^{-2}$ follows by differentiating its positive Laplace integral. A Stieltjes representation is

$$f(\lambda) = a/\lambda + b + \int_{[0, \infty)} \frac{\rho(dt)}{\lambda + t}, \quad a, b \geq 0, \quad \int \frac{\rho(dt)}{1+t} < \infty.$$

If this function is nonzero, either $a > 0$, $b > 0$, or ρ has positive mass on some bounded interval. In every case $\liminf_{\lambda \rightarrow \infty} \lambda f(\lambda) > 0$. For $(1+\lambda)^{-2}$ that limit is zero, a contradiction. Finally $p'(t) = (1-t)e^{-t} > 0$ for $0 < t < 1$, while $\bar{F}_\Gamma''(t) = (t-1)e^{-t} < 0$ there. These signs violate the respective complete-monotonicity inequalities. $\square$

Proposition 3.16 (Reverse size bias, equilibrium, and marked tilting). *Let μ be a probability on $[0, \infty)$ with finite positive mean m .*

1. *Its size-biased probability is $\beta(dt) = t\mu(dt)/m$. If $\mu(\{0\}) = 0$, then*

$$c = \int_{(0, \infty)} t^{-1} \beta(dt) = 1/m, \quad \mu(dt) = \frac{\beta(dt)}{ct}.$$

Conversely any probability β on $(0, \infty)$ with $0 < c < \infty$ in this formula has that unique positive inverse. In particular the positive size-bias inverse of μ_Γ is $\operatorname{Exp}(1)$.

2. *Its equilibrium law has canonical decreasing right-continuous density $q_E(t) = \mathbb{P}(X > t)/m$ for $t \geq 0$. If $\mathbb{P}(X > 0) = 1$, its inverse is*

$$m = 1/q_E(0+), \quad \mathbb{P}(X > t) = mq_E(t).$$

In particular the full equilibrium density $(1+t)e^{-t}/2$ identifies μ_Γ , without assuming a density for the candidate original lifetime.

3. For a probability μ on $\mathbb{R}$, a marked real tilt θ with $0 < M(\theta) = \int e^{\theta t} \mu(dt) < \infty$ gives $\mu_\theta(dt) = e^{\theta t} \mu(dt) / M(\theta)$ and has the inverse

$$\mu(dt) = \frac{e^{-\theta t} \mu_\theta(dt)}{\int e^{-\theta y} \mu_\theta(dy)}.$$

For μ_Γ , the admissible real tilts are $\theta < 1$, and

$$p_\theta(t) = (1 - \theta)^2 t e^{-(1-\theta)t}, \quad \kappa_n(T_\theta) = \frac{2(n-1)!}{(1-\theta)^n}.$$

Its mean is $2/(1-\theta)$ and variance $2/(1-\theta)^2$.

Allowing an atom at zero destroys the first two inverses. Forgetting the tilt mark leaves a rate ambiguity in the third.

Proof. In the first clause, integrating $1/t$ against the definition gives $c = \mu((0, \infty))/m = 1/m$. The inverse has total mass one and mean $1/c$ by direct integration, and size-biasing it returns β . For $\beta(dt) = p(t)dt$, one has $c = 1$, and division by t gives $e^{-t}dt$.

The tail integral identity $\int_0^\infty \mathbb{P}(X > t)dt = m$ normalizes the equilibrium density. Positivity of the original lifetime gives its right limit at zero as $1/m$; the displayed inverse therefore recovers the entire survival, including all positive atoms. Two right-continuous monotone representatives that agree almost everywhere agree everywhere, since a strict difference at one point would persist on a short interval to its right. Thus equality of equilibrium laws suffices. For the target density $q_E(0+) = 1/2$, the recovered survival is $(1+t)e^{-t}$, whose negative derivative is $p(t)$.

Multiplication of the tilt formula by $e^{-\theta t}$ and normalization proves the third inverse. The Gamma transform gives $M(\theta) = (1-\theta)^{-2}$ for $\theta < 1$ and divergence otherwise. Its tilted log MGF is $-2\log(1 - u/(1-\theta))$, proving the cumulants.

Every $r\delta_0 + (1-r)\text{Exp}(1)$, $0 \leq r < 1$, has the same size bias μ_Γ . Similarly $r\delta_0 + (1-r)\mu_\Gamma$ has the same equilibrium law for all $r < 1$, because its tail and mean are both multiplied by $1-r$. For the tilt boundary, a Gamma law of any rate $b > 0$, tilted by $\theta = b-r$, has the fixed output rate $r > 0$. Hence the output without its input tilt mark does not recover b . $\square$

Corollary 3.17 (Equilibrium fixed points). *A positive lifetime with finite mean m equals its equilibrium law if and only if it is exponential with rate $1/m$.*

Proof. Equality to the equilibrium law makes the original distribution absolutely continuous with density $\bar{F}/m$. Consequently $\bar{F}' = -\bar{F}/m$ almost everywhere and $\bar{F}(0) = 1$. Integration gives $\bar{F}(t) = e^{-t/m}$. This exponential law satisfies the same equation and mean by direct integration. $\square$

Theorem 3.18 (Rooted coefficients, Borel probabilities, and the inverse germ). *There is a unique formal power series $R(z) \in z\mathbb{R}[[z]]$ with $R = ze^R$, and for $n \geq k \geq 1$,*

$$[z^n]R(z)^k = \frac{k}{n} \frac{n^{n-k}}{(n-k)!}. \quad (22)$$

In particular

$$R(z) = \sum_{n \geq 1} \frac{n^{n-1}}{n!} z^n, \quad \text{radius}(R) = e^{-1},$$

and its compositional inverse germ is $z = te^{-t} = p(t)$. The marked Borel PGF and its probabilities are

$$B(s) = R(s/e), \quad b_n = \frac{e^{-n} n^{n-1}}{n!} \quad (n \geq 1), \quad B(s) = se^{B(s)-1}, \quad B(1) = 1. \quad (23)$$

Its inverse germ is $s = te^{1-t} = ep(t)$, with the factor e part of the marking. For k independent Borel variables their sum has probabilities

$$[z^n]B(s)^k = \frac{k}{n} e^{-n} \frac{n^{n-k}}{(n-k)!} \quad (n \geq k). \quad (24)$$

The complete rooted EGF or complete Borel PGF thus determines the inverse analytic germ. If a candidate global density is real analytic on the connected interval $(0, \infty)$ and agrees with that inverse germ near zero, it equals p everywhere. Alternatively te^{-t} specifies a canonical global extension of the germ.

Proof. Successive coefficient comparison in $R = ze^R$ determines each coefficient from preceding ones and gives a unique formal solution with linear coefficient one. The inverse function theorem applied to te^{-t} at zero gives a convergent solution germ, so its coefficients are those of the formal solution. For completeness, the residue change of variables $z = ue^{-u}$ gives

$$\begin{aligned} [z^n]R(z)^k &= \text{Res}_{u=0} u^{k-n-1} e^{nu} (1-u) du \\ &= [u^{n-k}]e^{nu} - [u^{n-k-1}]e^{nu} = \frac{k}{n} \frac{n^{n-k}}{(n-k)!}. \end{aligned}$$

When $n = k$, the second coefficient is zero and the same formula gives one. The coefficient ratio for $k = 1$ tends to e , since it is $(1 + 1/n)^{n-1}$, so the radius is e^{-1} . The inverse identity and Borel scaling follow by substitution.

For $0 \leq s < 1$, the positive-coefficient solution is the inverse on $[0, 1)$ of the strictly increasing map $t \mapsto te^{1-t}$ on $[0, 1)$. Indeed the local inverses agree, and continuation along this interval is possible because its derivative is $e^{1-t}(1-t) > 0$. Hence $B(s) \uparrow 1$ as $s \uparrow 1$. Monotone convergence of the nonnegative series proves $\sum b_n = 1$. Products of PGFs for independent sums and (22) give (24).

For labelled rooted trees, removing the root leaves an unordered set of rooted trees on disjoint sets of labels. The exponential formula gives precisely $R = ze^R$; hence their number on n labels is n^{n-1} . An ordered k -tuple of such components has EGF R^k ; an unordered set of k components has EGF $R^k/k!$. These conventions are part of the respective count statements. Finally equality of the inverse germs and the real-analytic identity theorem prove the global converse in its stated class. $\square$

Theorem 3.19 (Borel total size identifies a one-ancestor iid GW law). *In a Galton–Watson tree started from one ancestor, with independent identically distributed offspring counts and the usual rooted-tree construction fixed, the total size has law (23) if and only if the offspring law is $\text{Pois}(1)$. This determines the full tree law within that construction.*

Proof. Construct a countable independent family of offspring counts indexed by all finite words in positive integers, and retain a word as a vertex exactly when its ancestors and the indicated child indices are present. For Poisson offspring the offspring PGF is $\Phi(w) = e^{w-1}$. The extinction probability is the increasing limit of its iterates starting at zero, and therefore the least fixed point of Φ in $[0, 1]$. For $0 < w < 1$ one has $\log w < w - 1$, so $\Phi(w) > w$; also $\Phi(0) > 0$. Its only such fixed point is one. Extinction is therefore almost sure, and the tree, having finite offspring counts, has finite total size almost surely.

For any offspring PGF Φ in this iid construction, conditioning on the root offspring and using independence of its descendant trees gives $B(s) = s\Phi(B(s))$ for the total-size PGF when the total size is almost surely finite. For the forward Poisson construction this is $B = se^{B-1}$, whose formal coefficients are the Borel probabilities in Theorem 3.18. Conversely, if the observed total-size law is Borel, then its PGF satisfies both equations. The function B maps $(0, 1)$ onto $(0, 1)$, so division by s gives $\Phi(w) = e^{w-1}$ for every $0 < w < 1$. Analytic coefficient uniqueness identifies every offspring probability as $e^{-1}/n!$. The independent product construction then fixes the tree law. $\square$

Proposition 3.20 (Tree and continuation boundaries). *A complete Borel size law does not identify an arbitrary random rooted tree. A complete inverse germ does not identify an arbitrary smooth positive probability density on $(0, \infty)$, even with the canonical maximum, strict two-branch shape, and endpoint limits. Criticality or a leading Borel tail asymptotic alone does not identify the complete Borel law.*

Proof. Sample N with probabilities b_n and return either a rooted path with N vertices or a rooted star with N vertices. These laws have the same size, but on $N = 3$ their root degrees differ, and $b_3 > 0$. A Poisson GW tree also has positive probability of root degree two, which a path never has. Likewise the PGF product in (24) constructs independent components; the sizes of an independently designated forest need not be independent merely because component marginals have been specified.

Choose nonzero $h \in C_c^\infty((2, 3))$ with $\int h = 0$. For small nonzero ε , set $p_\varepsilon = p + \varepsilon h$. On the compact support both p and $-p'$ have positive minima, so positivity and strict decrease survive for small $|\varepsilon|$. Outside the support it equals p . The integral remains one, the complete germ at zero is unchanged, and the maximum at one and all endpoint limits remain canonical. Nevertheless the density is different globally.

The critical offspring law assigning probabilities $1/2, 1/2$ to zero and two children has mean one and is not Poisson; its trees have only odd total sizes. Finally Stirling’s formula gives $b_n \sim (2\pi)^{-1/2} n^{-3/2}$. Altering b_1, b_2 by opposite

sufficiently small amounts preserves positivity, total mass, and this exact tail asymptotic, while changing the probability law. $\square$

Theorem 3.21 (The calibrated maximum-entropy characterization). *Among Lebesgue probability densities f on $(0, \infty)$ satisfying*

$$\int t f(t) dt = 2, \quad \int f(t) \log t dt = 1 - \gamma_E, \quad \int f(t) |\log t| dt < \infty,$$

the unique maximizer of differential entropy is p , up to equality almost everywhere, and its entropy is $1 + \gamma_E$. Here the extended convention allows entropy $-\infty$ and defines it by the relative-entropy formula below; for finite entropy this agrees with $-\int f \log f$.

Proof. The target constraints follow from Theorem 3.2. Since $\log p = \log t - t$, every candidate has the finite cross entropy $\int f \log p = -1 - \gamma_E$. Put $z = f/p$ and define

$$D(f\|p) = \int_0^\infty p(t) \{z(t) \log z(t) - z(t) + 1\} dt \in [0, \infty].$$

The integrand is nonnegative and vanishes exactly when $z = 1$. Because both densities have mass one, this is the usual relative entropy, and $h(f) = 1 + \gamma_E - D(f\|p) \leq 1 + \gamma_E$. For p the divergence is zero; equality for a candidate forces $f = p$ almost everywhere. This proves both maximality and its exact converse. The two constraints alone do not identify p : use four disjoint compactly supported smooth bumps, solve the three linear orthogonality equations against $1, t, \log t$, and add a sufficiently small nonzero resulting perturbation to p . Positivity and all three equations survive, while the strict entropy inequality holds. $\square$

Theorem 3.22 (Unique probability convolution completion). *In the category of probabilities on $[0, \infty)$, there is a unique family $(\mu_r)_{r \geq 0}$ with $\mu_0 = \delta_0$, $\mu_{r+s} = \mu_r * \mu_s$, and $\mu_1 = \mu_\Gamma$. It is*

$$\mu_r = \text{Gamma}(2r, 1) \ (r > 0), \quad L_{\mu_r}(\lambda) = (1 + \lambda)^{-2r} \ (\lambda \geq 0). \quad (25)$$

Weak continuity in r is a conclusion, not an additional premise. The associated process with stationary independent increments is uniquely specified in finite-dimensional law once that process category and the starting value zero are supplied.

Proof. Euler's Gamma integral gives the displayed transforms, and their products prove the convolution equations by Laplace uniqueness. For uniqueness let (η_r) be another family. At fixed $\lambda \geq 0$, its Laplace transform is strictly positive and at most one, so $g(r) = -\log L_{\eta_r}(\lambda)$ is a nonnegative additive function of $r \geq 0$. It is monotone, since $g(s) - g(r) = g(s - r) \geq 0$ when $s \geq r$. Rational additivity and squeezing by rational times yield $g(r) = rg(1) = 2r \log(1 + \lambda)$ for every real $r \geq 0$. Another application of Laplace uniqueness gives (25). The continuity theorem for probability Laplace transforms now gives weak continuity, since

these transforms vary continuously and their pointwise limits are the displayed probability transforms.

For increasing times, take independent increments with the laws indexed by the successive time differences and form their partial sums. Convolution ensures consistency under inserting or deleting times; the probability extension theorem constructs the process law, and these prescribed products determine every finite-dimensional distribution. In the usual subordinator path convention one takes its right-continuous increasing version. The time-one marginal alone does not specify an independently supplied process: the process $X_r = rT$ with $T \sim \mu_\Gamma$ has that marginal and dependent increments, since two unit increments are the same nonconstant variable T . $\square$

Proposition 3.23 (Gamma Lévy measure, drift, and complete Bernstein structure). *The Gamma convolution semigroup has exponent and Lévy measure*

$$\psi(\lambda) = 2\log(1 + \lambda) = \int_0^\infty (1 - e^{-\lambda x}) \frac{2e^{-x}}{x} dx, \quad \nu(dx) = \frac{2e^{-x}}{x} dx. \quad (26)$$

Among probability subordinators with this exact Lévy measure, the following are equivalent and each identifies that semigroup: zero drift; $\psi(\lambda)/\lambda \rightarrow 0$; time-one support infimum zero; and time-one mean two. The Lévy measure alone leaves arbitrary nonnegative drift. Moreover

$$\frac{2e^{-x}}{x} = 2 \int_1^\infty e^{-sx} ds, \quad \log(1 + \lambda) = \int_1^\infty \frac{\lambda}{s(s + \lambda)} ds. \quad (27)$$

Thus ψ is a complete Bernstein function and $\psi'(\lambda) = 2/(1 + \lambda)$ is Stieltjes.

Proof. The integrability condition $\int (1 \wedge x) \nu(dx) < \infty$ follows from boundedness of $2e^{-x}$ near zero and its exponential tail after division by x . Differentiating the integral in (26) with respect to λ gives $2/(1 + \lambda)$ by dominated convergence on compact nonnegative λ intervals. Its value at zero is zero, so integration proves the exponent formula.

The Lévy–Khintchine representation for a subordinator is given in [1, Theorem 3.2, p. 22]. With this measure it is $k + d\lambda + 2\log(1 + \lambda)$, with killing $k \geq 0$ and drift $d \geq 0$. Probability mass one excludes killing and leaves $d\lambda + 2\log(1 + \lambda)$. Its time-one law is $d + T$, by multiplication of Laplace transforms and uniqueness; its support infimum is d , its mean is $d + 2$, and the asymptotic ratio of the exponent to λ is d . This proves all the equivalences and the drift counterexample. In particular the complete time-one law in Theorem 3.22 already forces zero drift.

The first identity in (27) is the elementary exponential integral; the second follows by integrating $1/s - 1/(s + \lambda)$ from one to infinity. Differentiation of the first on compact positive x intervals shows that the Lévy density is completely monotone. By definition this makes its Bernstein exponent complete. The derivative is the Stieltjes transform of $2\delta_1$, proving the last assertion. $\square$

Corollary 3.24 (Complete integer tails identify the Gamma exponent). *Let f be a Bernstein function. If, for some integer $N \geq 1$,*

$$f(n) = 2 \log(1 + n) \quad \text{for every integer } n \geq N,$$

then $f(\lambda) = 2 \log(1 + \lambda)$ for every $\lambda > 0$.

Proof. The target is Bernstein by Proposition 3.23. Apply the complete integer-tail uniqueness theorem 4.16 to f and this target. That theorem derives its finite compact moment measure from the Lévy–Khintchine representation and does not assume a separately given value at zero. $\square$

Proposition 3.25 (Explicit Gamma self-decomposition). *For $0 < c < 1$ there exists a nonnegative compound-Poisson residual Y_c , independent of a Gamma copy T' , such that $T \stackrel{d}{=} cT' + Y_c$. Its Lévy measure, total jump intensity, Laplace transform, and mass at zero are respectively*

$$\nu_c(dx) = \frac{2(e^{-x} - e^{-x/c})}{x} dx, \quad \nu_c((0, \infty)) = -2 \log c, \quad (28)$$

$$R_c(\lambda) = \left(\frac{1 + c\lambda}{1 + \lambda} \right)^2, \quad \mathbb{P}(Y_c = 0) = c^2. \quad (29)$$

In particular the Gamma law is infinitely divisible and self-decomposable. Those two general properties alone do not identify its shape or rate.

Proof. The density in (28) is nonnegative. Since $e^{-x} - e^{-x/c} = \int_1^{1/c} x e^{-sx} ds$, Tonelli gives its total mass as $2 \int_1^{1/c} ds/s = -2 \log c$. A Poisson number of independent positive jumps with normalized law $\nu_c/(-2 \log c)$ therefore constructs Y_c . Its log Laplace transform is $-\int (1 - e^{-\lambda x}) \nu_c(dx)$. Subtracting the two integrals in (26), with the change of variables for the second, gives

$$\int (1 - e^{-\lambda x}) \nu_c(dx) = 2 \log(1 + \lambda) - 2 \log(1 + c\lambda).$$

Exponentiation proves (29); the the mass at zero is the no-jump probability $e^{2 \log c} = c^2$. On a product space take an independent T' with law μ_Γ . The product of the Laplace transforms of cT' and Y_c is $(1 + \lambda)^{-2}$, proving the decomposition by uniqueness. Infinite divisibility also follows from the n -fold convolution of the time-1/ n law in Theorem 3.22. The identical construction with density $\alpha e^{-\beta x}/x$, for arbitrary $\alpha, \beta > 0$, proves both properties for every Gamma shape and rate, providing distinct laws with those same broad properties. $\square$

Theorem 3.26 (One linked residual determines a law on the entire real line). *Fix $0 \leq c < 1$. Let Y_c have the probability law whose Laplace transform is (29), equivalently the law of a sum of two independent variables with law $c\delta_0 + (1 - c) \text{Exp}(1)$. For an arbitrary Borel probability μ on $\mathbb{R}$, the following are equivalent:*

1. $\mu = \mu_\Gamma$;

2. there are independent variables X' and Y_c , with $X' \sim \mu$, such that $cX' + Y_c \sim \mu$;
3. the characteristic function of μ satisfies

$$\varphi(\xi) = \varphi(c\xi) \left(\frac{1 - ic\xi}{1 - i\xi} \right)^2 \quad (\xi \in \mathbb{R}).$$

No support, density, or moment assumption on μ is needed. The residual itself determines c within this family through $\mathbb{P}(Y_c = 0) = c^2$, so its scale may be recovered from the complete residual observation. The source-law linkage is essential; the contraction endpoint $c = 1$ does not identify μ .

Proof. A variable with law $c\delta_0 + (1 - c)\text{Exp}(1)$ has Laplace transform $c + (1 - c)/(1 + \lambda) = (1 + c\lambda)/(1 + \lambda)$. The sum construction proves existence of the residual and its transform, including $c = 0$. Both summands are nonnegative and have an atom of mass c at zero, so the sum has an atom of mass c^2 at zero. Its characteristic function is $((1 - ic\xi)/(1 - i\xi))^2$, by the same calculation.

Independence gives the factorization in the third clause from the second. Conversely that factorization and a product-space construction, followed by characteristic-function uniqueness, give the second clause. Iterating the factorization for any integer $n \geq 1$ telescopes:

$$\varphi(\xi) = \varphi(c^n \xi) \left(\frac{1 - ic^n \xi}{1 - i\xi} \right)^2.$$

The denominators are nonzero for real ξ . Since $c^n \xi \rightarrow 0$ and every probability characteristic function is continuous at zero with value one, letting $n \rightarrow \infty$ gives $\varphi(\xi) = (1 - i\xi)^{-2}$. Theorem 3.2 and characteristic-function uniqueness now identify μ on all of $\mathbb{R}$. For the converse, the Gamma characteristic function satisfies the factorization by direct cancellation, and independent variables with the required laws can be constructed on a product space. This also covers $c = 0$, where Y_0 is already Gamma.

Forgetting linkage permits the very same observed Y_c alongside an unrelated designated X with either Gamma shape two or Gamma shape three, so it identifies neither such designation. At $c = 1$ the residual is δ_0 and the factorization reduces to $\varphi(\xi) = \varphi(\xi)$ for every law. Thus contraction and the equation linking the original law to this precise residual are substantive identifying premises. Independence may be replaced by the exact factorization as stated, but that factorization does not follow for an arbitrary supplied coupling. $\square$

Remark 3.27 (Formalization). The probability development, including its measure-theoretic and analytic statements and counterexamples, is part of the complete Lean 4 formalization with no admitted proofs. The Bernstein integer-tail inverse used above is Theorem 4.16.

4 Stein characterization and the marked Laguerre realization

Throughout this section, $p(t) = te^{-t}$ is the density in the intrinsic decomposition of Σ , in the specified real coordinate $t > 0$. The complete identifying data below reconstruct Σ with their coordinate and operator context retained. The candidate classes for a coordinate probability, an abstract self-adjoint operator, and a semigroup mixing measure have their respective identifying equations. All Hilbert spaces are complex. We use Id for the identity operator, and reserve I for the scalar potential. The arguments give analytic proofs with explicit domains and candidate classes; numerical audits are not used to establish these universal statements.

Theorem 4.1 (The full-line weak Stein characterization). *Let P be a positive Borel probability measure on $\mathbb{R}$. Then*

$$\int_{\mathbb{R}} [tf'(t) + (2-t)f(t)] P(dt) = 0 \quad (f \in C_c^\infty(\mathbb{R})) \quad (30)$$

if and only if

$$P(dt) = \mathbf{1}_{(0,\infty)}(t)te^{-t} dt.$$

In particular, absolute continuity, positive support, absence of an atom at zero, and all finite moments are conclusions, not hypotheses.

Proof. The integrands in (30) have compact support, so no moment assumption is needed to state the identity. In distributions it says $(tP)' = (2-t)P$. On either open half-line put $Q = tP$. There

$$Q' = (2/t - 1)Q, \quad (e^t t^{-2} Q)' = 0.$$

A distribution with zero derivative on a connected interval is a constant distribution. Thus the restrictions of P have densities $C_+ te^{-t}$ on the positive half-line and $C_- |t|e^{-t}$ on the negative half-line, with $C_+, C_- \geq 0$. Finiteness of P forces $C_- = 0$. The only remaining possible part is $m\delta_0$, $m \geq 0$. The positive-half-line flux $t^2 e^{-t}$ tends to zero at the origin; its distributional derivative therefore has no atom there. Substitution of $m\delta_0$ into the equation gives $0 = 2m\delta_0$, so $m = 0$. Finally $\int_0^\infty te^{-t} dt = 1$ gives $C_+ = 1$. Conversely, integration by parts, using $(t^2 e^{-t})' = (2-t)te^{-t}$ and the zero endpoint flux, proves the displayed identity. Tests supported only in $(0, \infty)$ could not exclude mass on $(-\infty, 0]$; the full-line test class is essential to this support conclusion. $\square$

Proposition 4.2 (The integrable centered Stein kernel). *Fix the probability density p , and let τ be real measurable with $\tau p \in L_{\text{loc}}^1(0, \infty)$. The equation*

$$(\tau p)' = (2-t)p$$

in distributions is equivalent to

$$\tau(t) = t + C \frac{e^t}{t} \quad \text{almost everywhere,} \quad C \in \mathbb{R}. \quad (31)$$

The locally absolutely continuous representative of τp has both endpoint limits equal to C . Each of the following separately selects $\tau = t$ almost everywhere: zero flux at either endpoint; $\tau \in L^1(p dt)$; a finite expectation of τ under $p dt$; or the finite identity $\int \tau p = \text{Var}_p(t) = 2$. Nonnegativity alone permits exactly $C \geq 0$.

Proof. Write $q = tp = t^2 e^{-t}$. Since $q' = (2 - t)p$, the distribution $\tau p - q$ has derivative zero and is constant. This proves (31) and the assertions about its absolutely continuous representative. For $C \neq 0$, $q + C$ tends to C at infinity, so $\int_0^\infty |\tau|p = \int_0^\infty |q + C| = \infty$; its tail also has the constant sign of C , so $\int \tau p dt$ cannot be finite as a Lebesgue integral. For $C = 0$, integration by parts gives $\int tp = 2$, $\int t^2 p = 6$, and hence variance 2. As $q > 0$ and q tends to zero at both endpoints, $q + C \geq 0$ almost everywhere exactly when $C \geq 0$.

Every value of C satisfies the compact-test identity

$$\int \tau f' p dt = \int (t - 2) f p dt \quad (f \in C_c^\infty(0, \infty)),$$

because the added term is $C \int f' = 0$. Including $f(t) = t$, with both integrals required finite, instead gives $\int \tau p = 6 - 4 = 2$. This inverse problem fixes P and varies τ ; it is different from Theorem 4.1, which fixes the coefficient t and varies P . $\square$

Theorem 4.3 (Two marked probes and the Pearson realization). *In the conservative local second-order class*

$$Bf = -a(t)f'' - b(t)f', \quad t > 0,$$

the two differential-expression identities

$$B(2 - t) = 2 - t, \quad B(t^2/2 - 3t + 3) = t^2 - 6t + 6 \quad (32)$$

hold if and only if $a(t) = t$ and $b(t) = 2 - t$. The conclusion is pointwise or almost everywhere according to the meaning of the coefficient identities. If a positive, locally absolutely continuous probability density w is also required to obey the zero-current Pearson equation $(aw)' = bw$, then $w = p$. Thus these data determine the differential expression and its normalized Hilbert weight in this specified coordinate class.

Proof. The first probe gives $b = 2 - t$. The second then gives

$$-a - (2 - t)(t - 3) = t^2 - 6t + 6,$$

and consequently $a = t$. Substitution proves the converse. The Pearson equation reduces to $tw' = (1 - t)w$, so $(e^t w/t)' = 0$ locally and $w = Cte^{-t}$. Probability normalization gives $C = 1$. The probes are polynomials; they are identities of the expression, or identities on an explicitly enlarged domain containing those polynomials. They are not assertions that polynomials belong to $C_c^\infty(0, \infty)$. $\square$

Proposition 4.4 (The exact limits of the probe and stationary data). *The pair (32) is irredundant within the indicated coefficient class, even if the coefficients are smooth and $a > 0$. It does not identify an arbitrary self-adjoint operator outside that local class. The invariant density p alone does not identify a reversible dynamics or its time scale.*

Proof. With only the linear probe, set $b = 2 - t$ and choose any positive smooth a . With only the quadratic probe, take a nonzero $k \in C_c^\infty(0, \infty)$ and set

$$b = 2 - t + \varepsilon k(t), \quad a = t - \varepsilon k(t)(t - 3).$$

For sufficiently small $|\varepsilon|$, positivity holds on the compact support of k and elsewhere $a = t$; the quadratic identity holds identically while the linear one changes. For the larger self-adjoint category, use the operator and eigenbasis constructed below. Let U fix e_0, e_1, e_2 and every $e_n, n \geq 5$, and make a nontrivial rotation, with angle in $(0, \pi/2)$, on $\text{span}\{e_3, e_4\}$. Then $UAU^{-1} \neq A$, since the two rotated eigenvalues are distinct, but it preserves the constant vector and both probes. This example has self-adjointness; positivity preservation of the rotated semigroup is not imposed. Finally $cA, c > 0$, gives distinct conservative reversible semigroups with the same invariant $p dt$, as follows from the construction below. The centered integrable Stein selection fixes $c = 1$; invariance alone does not. $\square$

4.1 The actual self-adjoint domain and complete spectral resolution

Set

$$\mathcal{H} = L^2((0, \infty), p(t) dt), \quad q(t) = tp(t) = t^2 e^{-t}, \quad \langle f, g \rangle = \int_0^\infty f \bar{g} p dt.$$

In particular $\langle \cdot, \cdot \rangle$ is linear in its first argument. The minimal differential operator considered here is

$$A_0 f = -p^{-1}(qf')' = -tf'' - (2 - t)f', \quad D(A_0) = C_c^\infty(0, \infty). \quad (33)$$

Theorem 4.5 (Essential self-adjointness and the full domain). *The operator A_0 is densely defined, nonnegative and symmetric. Both singular endpoints are limit point, its deficiency indices are $(0, 0)$, and its unique self-adjoint extension is $A = \overline{A_0} = A_0^*$. Its exact domain is*

$$D(A) = \{f \in \mathcal{H} : f, qf' \in AC_{\text{loc}}(0, \infty), \quad -p^{-1}(qf')' \in \mathcal{H}\}. \quad (34)$$

Here the regularity refers to the indicated representatives of the L^2 classes. For every $f \in D(A)$,

$$\lim_{t \downarrow 0} q(t)f'(t) = \lim_{t \rightarrow \infty} q(t)f'(t) = 0. \quad (35)$$

These limits are consequences of the domain, not additional boundary conditions. In particular no finite boundary value $f(0)$ is required.

Proof. Smooth compactly supported functions are dense in this weighted L^2 space, because the weight is positive and smooth on each compact interior interval. Integration by parts gives

$$\langle A_0 f, g \rangle = \int_0^\infty q f' \overline{g'} dt,$$

which proves symmetry and nonnegativity. The definition of the adjoint says that $f \in D(A_0^*)$ precisely when there is an $h \in \mathcal{H}$ with $-(qf')' = ph$ in distributions. The right side is locally integrable. Taking a primitive, dividing by the positive smooth function q , and taking a second primitive shows that qf' and f have locally absolutely continuous representatives. Conversely those representatives permit integration by parts against each compact test. Thus the adjoint is exactly the maximal operator in (34).

To determine the endpoints, two independent zero-energy solutions are

$$u_1(t) = 1, \quad u_2(t) = \int_1^t \frac{e^s}{s^2} ds.$$

Expanding $e^s/s^2 = s^{-2} + s^{-1} + O(1)$ near zero and using l'Hôpital's rule at infinity gives

$$u_2(t) = -t^{-1} + \log t + O(1) \quad (t \downarrow 0), \quad u_2(t) \sim e^t/t^2 \quad (t \rightarrow \infty).$$

Consequently $|u_2|^2 p \sim t^{-1}$ at zero and $|u_2|^2 p \sim e^t/t^3$ at infinity. Neither is integrable. The constant solution is square integrable near each endpoint. The Weyl endpoint alternative therefore makes both endpoints limit point: an endpoint is limit circle precisely when all solutions at one, equivalently every, spectral parameter are square integrable there. The deficiency-index theorem gives one deficiency at each limit-circle endpoint and zero at each limit-point endpoint. Its hypotheses hold here because $p, q > 0$, p, q^{-1} are locally integrable, and the potential is zero; see Gesztesy–Littlejohn–Nichols [6, Theorems 3.6 and 3.8]. Their minimal domain is the closure of compactly supported maximal-domain elements. It has the same closure as C_c^∞ here: on compact interior intervals those elements are in H^2 , and smooth approximation converges in the graph norm. This verifies the application to the actual domain (33), including the critical logarithmic divergence at the origin. It gives $\overline{A_0} = A_0^*$ and deficiency indices $(0, 0)$. Nonnegativity passes to this closure.

For the flux statement put $h = Af$. Cauchy–Schwarz and $\int p = 1$ give $\int |h|p \leq \|h\|_{\mathcal{H}}$. Thus $(qf')' = -ph$ is globally integrable and qf' has finite limits at both endpoints. A nonzero limit ℓ at zero would yield $f(t) \sim -\ell/t$, by integrating $f' = (\ell + o(1))/q$; a nonzero limit at infinity would give $f(t) \sim \ell e^t/t^2$. The same weighted-square divergences exclude both. This proves (35); in fact

$$q(t)f'(t) = - \int_0^t h p ds = \int_t^\infty h p ds, \quad \int_0^\infty h p dt = 0.$$

To see why finiteness of $f(0)$ cannot be added, multiply $\log \log(1/t)$ by a smooth cutoff supported sufficiently near zero and equal to one there. This function is in

$\mathcal{H}$, and near zero its image under the differential expression is $O((t \log(1/t))^{-1})$. Its weighted square integral is finite, since $\int_0^\varepsilon dt/(t \log^2(1/t)) < \infty$. It therefore belongs to (34) and has no finite value at zero. $\square$

Proposition 4.6 (The closed form and the conservative semigroup). *The closed form of A is*

$$\mathcal{E}(f, g) = \int_0^\infty q f' \overline{g'} dt,$$

with the exact domain

$$D(A^{1/2}) = \left\{ f \in \mathcal{H} : f \in AC_{\text{loc}}(0, \infty), \quad \int_0^\infty q |f'|^2 dt < \infty \right\}. \quad (36)$$

There are no additional endpoint traces. The semigroup e^{-sA} , $s \geq 0$, is positivity preserving, preserves the constant function 1, and preserves the probability $p dt$.

Proof. Denote the displayed form domain by V . A Cauchy sequence in $\|f\|_V^2 = \|f\|_{\mathcal{H}}^2 + \int q |f'|^2$ is Cauchy in ordinary H^1 on every compact interior interval, where both weights are bounded above and below by positive constants. Its local weak derivative equals the weighted derivative limit, proving completeness. Truncate real and imaginary parts of $f \in V$ at height N . The weak chain rule and dominated convergence show convergence in V as $N \rightarrow \infty$. For bounded f , choose cutoffs which rise on $[\varepsilon, 2\varepsilon]$, fall on $[R, 2R]$, and have derivatives bounded respectively by C/ε and C/R . Their energy is bounded by

$$C\varepsilon + \frac{C}{R^2} \int_R^{2R} t^2 e^{-t} dt \rightarrow 0.$$

The two derivative errors in $\chi f - f$ are $(\chi - 1)f'$ and $\chi' f$; the first tends to zero by dominated convergence, the second by this estimate and boundedness of f . The $\mathcal{H}$ norm also tends to zero. On the resulting compact interior support, mollification converges in H^1 and hence in V . Thus C_c^∞ is a form core, and the displayed form is the closure of the minimal form. Its represented self-adjoint operator extends A_0 and must equal A by Theorem 4.5.

The weak chain rule shows that real normal contractions preserve V and decrease its energy. Equivalently, the variational resolvent is positivity preserving and maps $[0, 1]$ into $[0, 1]$, as follows by testing its equation with the negative part and with $(u - 1)^+$. The exponential resolvent limit gives the same Markov property for e^{-sA} . Finally $1 \in D(A)$ and $A1 = 0$, so $e^{-sA}1 = 1$. Self-adjointness gives $\int e^{-sA} f p dt = \langle e^{-sA} f, 1 \rangle = \langle f, e^{-sA} 1 \rangle = \int f p dt$. $\square$

Theorem 4.7 (The Laguerre basis and the exact integer spectrum). *Define the polynomials with their marked normalization by*

$$L_n^{(1)}(t) = \frac{e^t}{n!t} \frac{d^n}{dt^n} (e^{-t} t^{n+1}) = \sum_{k=0}^n (-1)^k \binom{n+1}{k+1} \frac{t^k}{k!}, \quad n \in \mathbb{N}_0. \quad (37)$$

Then $e_n = L_n^{(1)}/\sqrt{n+1}$ is a complete orthonormal basis of $\mathcal{H}$, each $e_n \in D(A)$, and $Ae_n = ne_n$. In particular the spectrum is exactly $\mathbb{N}_0$, each eigenvalue is simple, and the resolvent is compact. If $f = \sum c_n e_n$, then

$$\begin{aligned} D(A) &= \left\{ f : \sum_{n=0}^{\infty} n^2 |c_n|^2 < \infty \right\}, & Af &= \sum_{n=0}^{\infty} n c_n e_n, \\ D(A^{1/2}) &= \left\{ f : \sum_{n=0}^{\infty} n |c_n|^2 < \infty \right\}, & \mathcal{E}(f, f) &= \sum_{n=0}^{\infty} n |c_n|^2. \end{aligned} \quad (38)$$

Proof. Leibniz's formula gives the finite sum in (37). Its coefficients a_k satisfy $(k+1)(k+2)a_{k+1} + (n-k)a_k = 0$, so

$$t(L_n^{(1)})'' + (2-t)(L_n^{(1)})' + nL_n^{(1)} = 0.$$

Every polynomial belongs to the maximal domain: the polynomial and its polynomial image are in $\mathcal{H}$, and all local regularity conditions hold. These are therefore genuine eigenfunctions of A . For $m < n$, integrating the Rodrigues formula n times by parts gives $\int p L_n^{(1)} t^m dt = 0$. The boundary terms vanish at infinity by exponential decay and at zero because derivatives of $e^{-t} t^{n+1}$ of order at most $n-1$ vanish to order at least two. For $m = n$ the integral is $(-1)^n \Gamma(n+2)$. The leading coefficient $(-1)^n/n!$ then gives

$$\int_0^{\infty} L_n^{(1)} L_m^{(1)} p dt = (n+1) \delta_{nm}.$$

For completeness, let $g \in \mathcal{H}$ be orthogonal to every polynomial and define $F(z) = \int_0^{\infty} g(t) p(t) e^{zt} dt$. Cauchy-Schwarz gives locally uniform domination of this integral and of all its differentiated integrands on $\operatorname{Re} z < 1/2$. Thus F is holomorphic there, and $F^{(n)}(0) = 0$ for all n . The identity theorem gives $F = 0$ on that half-plane. The finite complex measure $g p dt$, pushed forward by $y = e^{-t}$ and regarded as a measure on $[0, 1]$, consequently has every moment zero, since $F(-k) = 0$ for $k \in \mathbb{N}_0$. Uniform polynomial approximation on $[0, 1]$ shows that it integrates every continuous function to zero; measure uniqueness gives $g = 0$. This proves completeness. The unitary coefficient map now takes A to multiplication by n on $\ell^2(\mathbb{N}_0)$, with exactly the displayed domain. The form domain follows by taking square roots. The diagonal entries $(n+\alpha)^{-1}$ of any resolvent, $\alpha > 0$, tend to zero, proving compactness and the asserted spectrum. $\square$

Theorem 4.8 (Complete marked Laguerre orthogonality identifies the law). *Let P be a positive Borel probability measure on $\mathbb{R}$ for which every polynomial is integrable. With the specific polynomials (37),*

$$\int_{\mathbb{R}} L_n^{(1)}(t) P(dt) = 0 \quad (n \geq 1)$$

holds if and only if $P(dt) = \mathbf{1}_{(0, \infty)} p(t) dt$. The positive support is again a conclusion.

Proof. Mass fixes moment zero. The nonzero leading coefficient $(-1)^n/n!$ means that the n th displayed equation determines moment n from all smaller moments. The preceding Rodrigues calculation proves these equations for $p dt$, whose moments are $(n+1)!$. Induction therefore gives $\int t^n P(dt) = (n+1)!$ for all $n \geq 0$. For $0 < c < 1$, monotone convergence in the nonnegative even-power series gives

$$\int \cosh(ct) P(dt) = \sum_{n=0}^{\infty} (2n+1) c^{2n} = \frac{1+c^2}{(1-c^2)^2} < \infty.$$

As $e^{c|t|} \leq 2 \cosh(ct)$, the bilateral moment-generating function is holomorphic on $|\operatorname{Re} z| < 1$. Its Taylor series near zero, determined by the moments, is $\sum_{n \geq 0} (n+1) z^n = (1-z)^{-2}$. The identity theorem gives equality to the Gamma moment-generating function throughout the strip, including the imaginary axis. Uniqueness of characteristic functions identifies P on all of $\mathbb{R}$. The converse was already proved by Rodrigues's formula. $\square$

4.2 Complete spectral data, kernel formulas, and their inverse category

Theorem 4.9 (Exact trace domains and reconstruction of the unitary class). *For the operator A above, the ordinary traces are*

$$Z_A(s) = \operatorname{Tr}(\operatorname{Id} + A)^{-s} = \zeta(s), \quad \operatorname{Re} s > 1, \quad (39)$$

$$\Theta_A(\tau) = \operatorname{Tr} e^{-\tau A} = \frac{1}{1 - e^{-\tau}}, \quad \operatorname{Re} \tau > 0. \quad (40)$$

These are precisely their trace-class domains. The powers use the real logarithms of the positive eigenvalues $n+1$; the zero eigenspace is included through this shift.

Let $B \geq 0$ be self-adjoint with compact resolvent. Each of the following complete data sets determines its eigenvalue multiset, including multiplicities, and hence its unitary equivalence class: all finite heat traces $\operatorname{Tr} e^{-\tau B}$, $\tau > 0$; all shifted zeta traces on a real half-line $s > s_0$ on which they are finite; or all integer shifted traces $\operatorname{Tr}(\operatorname{Id} + B)^{-k}$, $k \geq 2$, with $\operatorname{Tr}(\operatorname{Id} + B)^{-2} < \infty$. The coordinate law of an unspecified realization is not part of this unitary-class conclusion.

Proof. The basis of Theorem 4.7 gives diagonal entries $(n+1)^{-s}$ and $e^{-\tau n}$. Their singular values are respectively $(n+1)^{-\operatorname{Re} s}$ and $e^{-n \operatorname{Re} \tau}$. Their sums converge exactly on the stated half-planes and there give the Dirichlet and geometric series in (39)–(40). Analytic or meromorphic continuations outside these domains do not become ordinary operator traces.

For the inverse, compact resolvent gives an orthonormal eigenbasis, finite multiplicities, and eigenvalues tending to infinity if there are infinitely many. For any nonempty remaining multiset let its least eigenvalue be λ_0 , of multiplicity m_0 . If $T(\tau)$ is its remaining heat trace, then

$$\lambda_0 = - \lim_{\tau \rightarrow \infty} \frac{\log T(\tau)}{\tau}, \quad m_0 = \lim_{\tau \rightarrow \infty} e^{\tau \lambda_0} T(\tau).$$

Indeed, after this rescaling the least-eigenvalue terms equal m_0 , and the other terms tend to zero. Finiteness at a fixed positive time supplies a summable dominating sequence for all larger times. Subtract $m_0 e^{-\tau \lambda_0}$ and iterate. The identically zero remainder detects the end of a finite spectrum, including the empty Hilbert space. For zeta data write $r_j = 1 + \lambda_j$. The analogous formulas are

$$r_0 = \lim_{s \rightarrow \infty} Z(s)^{-1/s}, \quad m_0 = \lim_{s \rightarrow \infty} r_0^s Z(s).$$

Finiteness at one exponent again dominates the rescaled tails. Subtracting $m_0 r_0^{-s}$ and repeating recovers every eigenvalue.

For integer data set $x_j = (1 + \lambda_j)^{-1}$ and form the finite positive measure

$$\eta_B = \sum_j x_j^2 \delta_{x_j} \quad \text{on } [0, 1], \quad \int x^n \eta_B(dx) = Z_B(n+2), \quad n \geq 0.$$

Two such measures with the same moments agree on polynomials and then on all continuous functions by uniform polynomial approximation. They are therefore equal. Every nonzero atom at x has mass $m x^2$, recovering the integer multiplicity m and eigenvalue $x^{-1} - 1$. There is no atom at zero from any finite eigenvalue. This is the compact Hausdorff moment argument; it uses no interpolation from the integer values. Finally an eigenvalue-preserving bijection of orthonormal eigenbases is an intertwining unitary, with the weighted coefficient domains carried to one another. $\square$

Proposition 4.10 (Compact resolvent does not supply finite trace data). *There is a nonnegative diagonal self-adjoint operator with compact resolvent for which neither a heat trace nor a positive shifted zeta trace is finite at any positive parameter.*

Proof. On $\ell^2(\mathbb{N})$ take eigenvalues $\lambda_j = \log \log(j + e^e)$, with domain $\{c : \sum \lambda_j^2 |c_j|^2 < \infty\}$. They tend to infinity, so $(\text{Id} + B)^{-1}$ is compact. For every $\tau > 0$, $(\log(j + e^e))^{-\tau}$ eventually exceeds $1/j$, and for every $s > 0$, $(1 + \log \log(j + e^e))^{-s}$ eventually exceeds $1/j$. Both trace series diverge. Thus the convergence hypotheses in Theorem 4.9 cannot be inferred from compactness alone. $\square$

Theorem 4.11 (The complete heat kernel relative to the invariant measure). *For $\tau > 0$, set $r = e^{-\tau}$. The kernel of $e^{-\tau A}$ relative to $p(y) dy$ is*

$$K_\tau(x, y) = \sum_{n=0}^{\infty} \frac{r^n}{n+1} L_n^{(1)}(x) L_n^{(1)}(y) \tag{41}$$

$$= \frac{\exp[-r(x+y)/(1-r)]}{(1-r)\sqrt{rxy}} I_1\left(\frac{2\sqrt{rxy}}{1-r}\right), \quad x, y > 0. \tag{42}$$

Here I_1 is the modified Bessel function, specified by $I_1(z) = \sum_{m \geq 0} (z/2)^{2m+1} / (m!(m+1)!)$. The series converges locally uniformly in $x, y > 0$ and in $L^2(p(x)p(y) dx dy)$. The kernel is positive and symmetric, $\int K_\tau(x, y) p(y) dy = 1$, and it tends to 1 as $\tau \rightarrow \infty$. The Lebesgue transition density is $K_\tau(x, y) p(y)$.

Proof. Here is a direct verification of the normalization and the closed formula, which is the parameter-one Hille–Hardy identity [7, Eq. 18.18.27]. The finite coefficients in (37) give the generating function

$$G_z(y) = \sum_{n \geq 0} L_n^{(1)}(y) z^n = (1-z)^{-2} \exp[-yz/(1-z)], \quad |z| < 1.$$

Let $\tilde{K}$ be the right side of (42). Expanding I_1 gives the nonnegative series

$$\tilde{K}(x, y) = e^{-r(x+y)/(1-r)} \sum_{m=0}^{\infty} \frac{(rxy)^m}{m!(m+1)!(1-r)^{2m+2}}.$$

For real z sufficiently near zero, put $c = (1-rz)/((1-r)(1-z)) > 0$. Termwise gamma integration, justified by positivity, gives

$$\begin{aligned} \int_0^{\infty} \tilde{K}(x, y) G_z(y) p(y) dy &= \frac{e^{-rx/(1-r)}}{(1-z)^2(1-r)^2 c^2} \exp\left(\frac{rx}{(1-r)^2 c}\right) \\ &= (1-rz)^{-2} \exp[-rxz/(1-rz)] = G_{rz}(x). \end{aligned}$$

In particular $z = 0$ gives row mass one. The kernel is symmetric and positive, so Jensen’s inequality and row mass one make its integral operator a contraction on $\mathcal{H}$. For z in a sufficiently small complex neighborhood of zero, the same exponentially dominated integral is holomorphic. Differentiating at zero proves that its operator maps $L_n^{(1)}$ to $r^n L_n^{(1)}$. Completeness of the basis identifies it with $e^{-\tau A}$. The spectral series is its L^2 kernel, since $\sum r^{2n} < \infty$. Cauchy’s coefficient estimate for G_z , on any circle $|z| = R$ with $\sqrt{r} < R < 1$, bounds its summands uniformly on compact sets by a convergent geometric sequence. Thus this series is continuous and locally uniformly convergent. Its L^2 equality with the continuous closed formula implies their pointwise equality on $(0, \infty)^2$. Finally the $m = 0$ Bessel term and the convergent series give $K_{\tau}(x, y) \rightarrow 1$ as $r \downarrow 0$. The definition of an integral kernel relative to $p(y) dy$ proves the assertion about the Lebesgue density. $\square$

Proposition 4.12 (Resolvents, their kernels, and the operator inverse). *For $\alpha > 0$, the kernels relative to $p(y) dy$ of $(A + \alpha \text{Id})^{-1}$ and $(A + \alpha \text{Id})^{-2}$ are respectively*

$$\begin{aligned} R_{\alpha}(x, y) &= \int_0^{\infty} e^{-\alpha\tau} K_{\tau}(x, y) d\tau, \\ S_{\alpha}(x, y) &= \int_0^{\infty} \tau e^{-\alpha\tau} K_{\tau}(x, y) d\tau. \end{aligned}$$

These identities hold as L^2 kernels, and the nonnegative integrals are finite almost everywhere. The first resolvent is Hilbert–Schmidt but is not trace class. The second is trace class and

$$\text{Tr}(A + \alpha \text{Id})^{-2} = \zeta(2, \alpha),$$

where $\zeta(s, \alpha) = \sum_{n=0}^{\infty} (n + \alpha)^{-s}$ for $\operatorname{Re} s > 1$. For $\alpha, \beta > 0$,

$$\operatorname{Tr}[(A + \alpha \operatorname{Id})^{-1} - (A + \beta \operatorname{Id})^{-1}] = \psi(\beta) - \psi(\alpha), \quad (43)$$

with $\psi = \Gamma'/\Gamma$. A complete resolvent operator, with its Hilbert realization and shift retained, determines A and its domain. A full marked heat operator at any fixed positive time also determines A .

Proof. On each eigenvector the Laplace integrals are $\int_0^{\infty} e^{-(n+\alpha)\tau} d\tau = (n+\alpha)^{-1}$ and $\int_0^{\infty} \tau e^{-(n+\alpha)\tau} d\tau = (n+\alpha)^{-2}$. The corresponding strong operator integrals converge, and their kernel integrals converge in L^2 because

$$\|K_{\tau}\|_{L^2(p \otimes p)} = (1 - e^{-2\tau})^{-1/2},$$

which is $O(\tau^{-1/2})$ near zero and bounded at infinity. The exponential weights are therefore integrable in this norm. Positivity and Tonelli identify these L^2 limits with the nonnegative pointwise integrals almost everywhere. The singular-value series for the resolvents are $\sum (n + \alpha)^{-1} = \infty$ and $\sum (n + \alpha)^{-2} < \infty$, proving the ideal statements. The difference series is absolutely summable and equals

$$\sum_{n=0}^{\infty} \left(\frac{1}{n + \alpha} - \frac{1}{n + \beta} \right) = \psi(\beta) - \psi(\alpha),$$

as follows by subtracting the convergent digamma expansions $\psi(a) = -\gamma_E + \sum_{n \geq 0} ((n+1)^{-1} - (n+a)^{-1})$. This also fixes the sign in (43).

For $R = (A + \alpha \operatorname{Id})^{-1}$ one has $D(A) = \operatorname{Ran} R$ and $A(Rh) = h - \alpha Rh$ for every $h \in \mathcal{H}$. Thus the complete operator recovers the generator, whereas a scalar resolvent trace discards its eigenvectors and coordinate realization. For $T = e^{-\tau A}$ the Borel functional calculus similarly gives $A = -\tau^{-1} \log T$, with its spectral domain; zero is not an eigenvalue of T , and its possible accumulation there is handled by this unbounded functional calculus. Consequently a full kernel with its reference measure retained recovers the corresponding operator. $\square$

Proposition 4.13 (Zeta and Fredholm determinants with their conventions). *For $\alpha > 0$, define the zeta determinant using the meromorphic continuation of the Hurwitz zeta function. Then*

$$\det_{\zeta}(A + \alpha \operatorname{Id}) = \exp[-\partial_s \zeta(0, \alpha)] = \frac{\sqrt{2\pi}}{\Gamma(\alpha)}. \quad (44)$$

In particular $\det_{\zeta}(\operatorname{Id} + A) = \det'_{\zeta}(A) = \sqrt{2\pi}$, where the prime omits the zero mode. For every $z \in \mathbb{C}$ one has the entire-function identities

$$\det(\operatorname{Id} + z(\operatorname{Id} + A)^{-2}) = \prod_{k=1}^{\infty} (1 + z/k^2) = \frac{\sinh(\pi\sqrt{z})}{\pi\sqrt{z}}, \quad (45)$$

$$\det_2(\operatorname{Id} + z(\operatorname{Id} + A)^{-1}) = \prod_{k=1}^{\infty} (1 + z/k) e^{-z/k} = \frac{e^{-\gamma_E z}}{\Gamma(1 + z)}. \quad (46)$$

The first determinant is the ordinary trace-class Fredholm determinant; the second is the Hilbert–Schmidt regularized determinant. The square-root quotient denotes its entire power series, with value 1 at zero. Its value is independent of the square-root choice. The full zero multiset of either displayed determinant identifies the eigenvalue multiset of A in the corresponding nonnegative, shifted determinant category.

Proof. The spectral series for $A + \alpha \text{Id}$ is the Hurwitz zeta series on $\text{Re } s > 1$. Its continuation is regular at zero and obeys $\partial_s \zeta(0, \alpha) = \log \Gamma(\alpha) - \frac{1}{2} \log(2\pi)$; see [7, Eq. 25.11.18]. This gives (44). The positive eigenvalues of A and all eigenvalues of $\text{Id} + A$ are both $1, 2, \dots$, which proves the two special determinant values with the stated zero-mode convention. The eigenvalues of $(\text{Id} + A)^{-2}$ are k^{-2} , a summable sequence; those of $(\text{Id} + A)^{-1}$ are k^{-1} , whose squares are summable. The determinant definitions therefore give the two products, locally uniformly in z . The identities [7, Eqs. 4.36.1 and 5.8.2] evaluate them as shown. Evenness of $\sinh w/w$ proves the branch statement. For a nonnegative operator B with the respective Schatten-class shifted inverse, a nonzero eigenvalue κ of that inverse produces a zero at $-1/\kappa$ with its multiplicity. The maps $\lambda \mapsto (1 + \lambda)^{-2}$ and $\lambda \mapsto (1 + \lambda)^{-1}$ are injective on $[0, \infty)$. The shift, exponent, and regularization convention therefore give the claimed inverse from the entire zero multiset. $\square$

Theorem 4.14 (Finite familiar spectral invariants do not determine the spectrum). *Fix any finite list of values of the continued shifted zeta function at nonzero real regular points, optionally its finite part at $s = 1$ and its zeta determinant. There is a nonconstant local family of nonnegative self-adjoint diagonal operators, differing from A in only finitely many positive eigenvalues, that retains all these specified invariants. Their zeta functions also retain every pole and principal part, and the value at zero. Thus this finite list, including a single zeta determinant as a special case, does not identify the complete spectrum.*

Proof. Vary N distinct shifted eigenvalues $r_j = 1 + \lambda_j > 1$ and fix the others. The correction

$$\Delta Z(s) = \sum_{j=1}^N (r_j^{-s} - r_{j,0}^{-s})$$

is entire, leaves every principal part unchanged, and has $\Delta Z(0) = 0$. Preserving a value at a nonzero real s_i imposes the equation $\sum r_j^{-s_i} = \sum r_{j,0}^{-s_i}$. Preserving the finite part at one imposes this same equation with $s_i = 1$. Preserving the determinant imposes $\sum \log r_j = \sum \log r_{j,0}$. After duplicate equations are removed, let their number be m . The gradient rows are nonzero multiples of $r_j^{-s_i-1}$, and, if present, the additional row r_j^{-1} . Their exponents are distinct.

These rows have rank m at distinct positive r_j whenever $N \geq m$. Indeed, put $u_j = \log r_j$. A nonzero linear combination of m exponentials with distinct real exponents has at most $m - 1$ distinct real zeros. Inductively, divide by its smallest exponential and differentiate; Rolle’s theorem would otherwise give at least $m - 1$ zeros for a nonzero combination of $m - 1$ exponentials. The single-exponential case starts the induction. Thus a nonzero row combination cannot vanish at m

distinct u_j . Choose $N > m$. The implicit-function theorem gives a local level manifold of dimension $N - m > 0$ through the original shifted eigenvalues. In small disjoint neighborhoods they remain greater than one and distinct; changes in this manifold cannot merely permute the spectrum. The fixed infinite tail preserves compact resolvent and every trace convergence domain.

For an explicit simultaneous example replace 2, 3, 4 by a, b, c , where

$$a = 2 + \varepsilon, \quad b + c = 7 - \varepsilon, \quad bc = \frac{24}{2 + \varepsilon}.$$

For sufficiently small nonzero ε , the quadratic roots b, c are real, distinct, greater than one, and close to 3, 4. The sum remains 9 and the product remains 24. Consequently $Z(0)$, $Z(-1)$, $Z'(0)$, the residue at one and the zeta determinant are unchanged, while the full spectrum changes. All eigenvalues outside this triple, including the zero eigenvalue, remain fixed. A determinant alone already permits the two-value perturbation $2 \mapsto 2c$, $3 \mapsto 3/c$ for c near 1. These countermodels concern the specified familiar invariants; they make no claim about an arbitrarily encoded finite datum. $\square$

4.3 The fixed mixing recipe and its exact unknown-measure inverse

Theorem 4.15 (Existence and uniqueness of the linked mixing measure). *Let $B \geq 0$ be a self-adjoint operator on a Hilbert space. The fixed function $J(\lambda) = (1 + \lambda)^{-2}$ always has the strong-operator mixing representation*

$$\int_0^\infty e^{-sB} s e^{-s} ds = (\text{Id} + B)^{-2}. \quad (47)$$

If B has a nonzero eigenvector for every $n \in \mathbb{N}_0$, then the unknown finite complex Borel measure ν on $[0, \infty)$ in

$$\int_{[0, \infty)} e^{-sB} \nu(ds) = (\text{Id} + B)^{-2} \quad (48)$$

is uniquely $\nu(ds) = s e^{-s} ds$. Finite total variation is understood for a complex measure. A single eigenvector at every integer suffices; multiplicities and additional spectrum do not affect the conclusion.

There is also a full-line formulation without initially defining negative-time operators. If ν is a finite positive Borel measure on $\mathbb{R}$ and its finite scalar integer integrals satisfy

$$\int_{\mathbb{R}} e^{-ns} \nu(ds) = (n + 1)^{-2} \quad (n \in \mathbb{N}_0), \quad (49)$$

then $\nu(ds) = \mathbf{1}_{(0, \infty)} s e^{-s} ds$. In this formulation mass, positive support and absence of an atom at zero are conclusions of the supplied samples.

Proof. The scalar gamma integral gives, for every $\lambda \geq 0$,

$$\int_0^\infty s e^{-(1+\lambda)s} ds = (1+\lambda)^{-2}.$$

The semigroup is a contraction, so each vector-valued integral is absolutely integrable. Integration against the spectral measure, justified by bounded domination, proves (47) for every nonnegative self-adjoint B .

For uniqueness, apply (48) to a nonzero n -eigenvector. This gives all scalar equations (49) on $[0, \infty)$. Under $y = e^{-s}$, their left sides are the moments of the finite complex pushforward measure on $(0, 1]$, extended by zero to $[0, 1]$. The finite positive measure with density $-\log y$ on $(0, 1)$ has these same moments, since

$$\int_0^1 y^n (-\log y) dy = (n+1)^{-2}.$$

Uniform polynomial approximation on $[0, 1]$ identifies the two finite measures. Undoing the measurable bijection $s = -\log y$ gives $\nu(ds) = s e^{-s} ds$. In particular the zeroth equation recovers its mass; its lack of an atom at $s = 0$ follows as well.

For a positive measure initially on all of $\mathbb{R}$, mass on $(-\infty, -\varepsilon]$ would imply

$$(n+1)^{-2} \geq e^{n\varepsilon} \nu((-\infty, -\varepsilon])$$

for every n , which is impossible for positive mass. Taking the union over $\varepsilon = 1/k$ proves support in $[0, \infty)$. The preceding compact moment proof then applies. No integral of unbounded negative-time operators was assumed in this support step. $\square$

Theorem 4.16 (A Bernstein function is determined by every integer tail). *Let f, g be Bernstein functions, in their usual representation category*

$$f(\lambda) = k_f + d_f \lambda + \int_{(0, \infty)} (1 - e^{-\lambda s}) \Pi_f(ds), \quad \lambda \geq 0, \quad (50)$$

where $k_f, d_f \geq 0$, Π_f is positive, and $\int (1 \wedge s) \Pi_f(ds) < \infty$, and likewise for g . For any fixed $N \in \mathbb{N}_0$, equality $f(n) = g(n)$ for every integer $n \geq N$ implies $f = g$ on $[0, \infty)$, including equality of killing, drift and Lévy measure. In particular the samples

$$f(n) = 2 \log(1+n) \quad (n \geq N)$$

identify the complete exponent $f(\lambda) = 2 \log(1+\lambda)$, with $k_f = d_f = 0$ and $\Pi_f(ds) = 2e^{-s} ds/s$. Knowledge of $f(A)$ on the marked integer eigenspaces supplies these samples. This inverse does not extend to arbitrary smooth functions.

The integer-tail uniqueness principle for Bernstein functions also appears in Aguech and Jedidi, Corollary 4 [2]. The proof below is independent and additionally records the recovery of the Lévy–Khinchine data in the present notation.

Proof. The finite positive measure

$$\rho_f = d_f \delta_1 + (s \mapsto e^{-s})_{\#} [(1 - e^{-s}) \Pi_f(ds)] \quad \text{on } [0, 1]$$

has no atom at zero. Its finiteness follows from $1 - e^{-s} \leq 1 \wedge s$. Direct subtraction of (50) gives

$$f(n+1) - f(n) = \int_0^1 y^n \rho_f(dy), \quad n \geq 0.$$

Equality of the integer tails therefore identifies all moments of $y^N \rho_f$ and $y^N \rho_g$. Compact moment uniqueness gives $y^N \rho_f = y^N \rho_g$. Since the weight is strictly positive on $(0, 1]$, division on $[1/j, 1]$ and passage to their increasing union give equality there. The absence of atoms at zero gives $\rho_f = \rho_g$ on $[0, 1]$. The atom at one recovers the drift $d_f = d_g$. The restriction to $(0, 1)$, the inverse coordinate $s = -\log y$, and division by the strictly positive weight $1 - e^{-s}$ recover $\Pi_f = \Pi_g$. With these fixed, a single common value at N recovers $k_f = k_g$; this also works when $N > 0$.

For the displayed target measure, the Lévy integrability condition holds at both endpoints. Its exponent is zero at $\lambda = 0$ and has derivative

$$\frac{d}{d\lambda} \int_0^\infty (1 - e^{-\lambda s}) \frac{2e^{-s}}{s} ds = 2 \int_0^\infty e^{-(1+\lambda)s} ds = \frac{2}{1+\lambda}.$$

Thus it equals $2 \log(1 + \lambda)$, proving the target assertion. Functional calculus gives $f(A)e_n = f(n)e_n$ and

$$D(f(A)) = \left\{ \sum c_n e_n : \sum |f(n)|^2 |c_n|^2 < \infty \right\}.$$

Consequently equality on the marked integer eigenspaces has exactly the required scalar meaning. For arbitrary smooth functions the nonzero perturbation $\varepsilon \sin^2(\pi\lambda)$ is invisible at every nonnegative integer; adding it to the nonnegative target also preserves nonnegativity. It therefore gives a distinct scalar function with the same functional calculus on A , outside the Bernstein inverse category just proved. $\square$

Proposition 4.17 (Two different functional-calculus inverse questions). *For a fixed strictly increasing Bernstein function f , the full self-adjoint operator $f(B)$ determines a nonnegative self-adjoint B . For the canonical A , the complete operator $2 \log(\text{Id} + A)$ is equivalently determined by $J(A) = (\text{Id} + A)^{-2}$. These statements retain the known scalar function f or J .*

Proof. Strict monotonicity and continuity give a Borel inverse on the range of f . The Borel functional calculus gives $B = f^{-1}(f(B))$, with its exact spectral domain. If an endpoint of the range is a spectral limit, the spectral projection at that unattained endpoint is zero, so the unbounded inverse is still well-defined. In the indicated special case, $J(A)$ has eigenvalues $(1+n)^{-2} > 0$, and $-\log J(A) = 2 \log(\text{Id} + A)$ on its functional-calculus domain. Inverting a known function of an unknown operator is thus different from Theorem 4.16, which identifies an unknown function from its values on a known integer spectrum. $\square$

Theorem 4.18 (The Gamma shapes two and three retain all spectral and mixing data). *For $\alpha \in \{2, 3\}$ set*

$$p_\alpha(t) = \frac{t^{\alpha-1}e^{-t}}{\Gamma(\alpha)}, \quad \mathcal{H}_\alpha = L^2((0, \infty), p_\alpha(t) dt),$$

and let A_α be the closure of

$$-tf'' - (\alpha - t)f', \quad D = C_c^\infty(0, \infty).$$

Both endpoints are limit point in both cases. The operators have the same simple complete spectrum $\mathbb{N}_0$ and are intertwined by a unitary carrying 1 to 1. Therefore all their heat traces, shifted zeta functions, resolvent trace functions and spectral determinants agree, with the same conventions and domains. Both satisfy (48) with the same unique mixing measure $\nu_2(ds) = se^{-s} ds$. Their invariant coordinate laws nevertheless differ: they are $p_2 dt$ and $p_3 dt$, with means 2 and 3 respectively.

Proof. The divergence-form coefficients are $q_\alpha = tp_\alpha$. Apart from an irrelevant nonzero constant, the second zero-energy solution is $\int_1^t e^s s^{-\alpha} ds$. At zero it is asymptotic to $-t^{1-\alpha}/(\alpha - 1)$; its weighted square is asymptotic to a positive multiple of $t^{1-\alpha}$, nonintegrable for both $\alpha = 2$ and 3. At infinity the solution is asymptotic to e^t/t^α , whose weighted square is a positive multiple of $e^t/t^{\alpha+1}$, also nonintegrable. The constant solution is square integrable at both ends. The maximal-domain and endpoint arguments of Theorem 4.5 therefore apply to each, giving their unique self-adjoint closures. The form argument also applies, since the lower-cutoff energy is $O(\varepsilon^{\alpha-1})$, proving conservativity and invariance of $p_\alpha dt$.

The Rodrigues polynomials

$$L_n^{(\alpha-1)}(t) = \frac{t^{1-\alpha}e^t}{n!} \frac{d^n}{dt^n}(e^{-t}t^{n+\alpha-1})$$

satisfy $A_\alpha L_n^{(\alpha-1)} = nL_n^{(\alpha-1)}$ because their coefficients obey $(k+1)(k+\alpha)a_{k+1} + (n-k)a_k = 0$. Integration by parts gives squared norm $\Gamma(n+\alpha)/(n!\Gamma(\alpha))$ in $\mathcal{H}_\alpha$. The weighted Laplace-transform proof of completeness applies without change: p_α has the same exponential tail and all polynomial moments. Hence

$$e_{n,\alpha} = \sqrt{\frac{n!\Gamma(\alpha)}{\Gamma(n+\alpha)}} L_n^{(\alpha-1)}$$

is a complete orthonormal eigenbasis, with $e_{0,\alpha} = 1$. The unitary $Ue_{n,2} = e_{n,3}$ intertwines the two operators and fixes the distinguished constant vectors. Every listed scalar spectral invariant therefore agrees. Theorem 4.15 gives the common mixing identity and its mixing-measure uniqueness. Finally gamma integration gives $\int tp_\alpha(t) dt = \alpha$, proving that the coordinate probabilities differ despite all those agreements. $\square$

Corollary 4.19 (Canonical mixing construction is not unmarked coordinate recovery). *The fixed exponent-two recipe J constructs the $\text{Gamma}(2, 1)$ mixing measure for every nonnegative self-adjoint operator. Integer eigenvalues make that mixing measure uniquely recoverable from the restricted operator identity. They do not identify the stationary coordinate law of an arbitrary realization, even when the constant vector is retained. To identify that candidate law one must retain either its explicit equality to the unknown mixing measure in (48), the local Pearson realization of Theorem 4.3, or equivalent coordinate information.*

Proof. Existence and mixing uniqueness are Theorem 4.15; the two realizations in Theorem 4.18 are a counterexample to the unmarked coordinate inverse, even under the conjunction of all the stated spectral and mixing data. For contrast, a fully marked package $(\mathcal{H}, A, 1, M_t)$ retains the multiplication operator $M_t f(t) = t f(t)$ on $\{f : t f \in \mathcal{H}\}$, and recovers its coordinate probability by

$$P(E) = \langle \mathbf{1}_E(M_t)1, 1 \rangle$$

for every Borel E . A unitary intertwining this multiplication operator as well as fixing 1 would preserve that probability, so the preceding unitary cannot do so. Furthermore, the different retained recipe $J_a(\lambda) = (1 + \lambda)^{-a}$, $a > 0$, has mixing measure $s^{a-1} e^{-s} ds / \Gamma(a)$ by the same gamma integral. Spectral data do not choose this exponent.

In particular let F forget a stationary coordinate realization and retain its spectrum, and let G_J construct the canonical $\text{Gamma}(2, 1)$ realization using the fixed recipe. Then

$$G_J(F(p_3, A_3)) = (p_2, A_2) \neq (p_3, A_3).$$

Thus this construction is not a left inverse to forgetting on the class of stationary coordinate realizations. An identification graph can use the linked unknown-measure theorem or retain the fixed coordinate template in its nodes; a cycle through an unmarked spectrum and canonical selection alone does not prove identification of its original candidate. $\square$

5 Complete characteristic series and topology

The complete characteristic series below recover the intrinsic decomposition of Σ . The coordinate in every formal statement is marked. Unless otherwise specified, coefficients lie in a field K of characteristic zero. The same identities hold in a commutative $\mathbb{Q}$ -algebra when every scalar used as a denominator is a unit. For a formal series, $[u^n]$ denotes coefficient extraction and Res_u is the coefficient of u^{-1} in a Laurent differential. A quotient such as $u/(1 - e^{-u})$ means the inverse of the unit $(1 - e^{-u})/u$; no inverse of the nonunit u is being asserted.

Theorem 5.1 (The normalized Todd tower and all its twists). *There is a unique series $Q(u) = \sum_{j \geq 0} q_j u^j$ satisfying*

$$q_0 = 1, \quad [u^n]Q(u)^{n+1} = 1 \quad (n \geq 1).$$

It is

$$Q_T(u) = \frac{u}{1 - e^{-u}}.$$

Equivalently, one may impose the coefficient equations for every $n \geq 0$, since their degree-zero equation is exactly $q_0 = 1$. For this series, for every $n \geq 0$,

$$[u^n]e^{ku}Q_T(u)^{n+1} = \binom{n+k}{n} = \frac{(k+1)(k+2)\cdots(k+n)}{n!}. \quad (51)$$

This is a polynomial identity in k over $\mathbb{Q}$, including the empty-product convention for $n = 0$, and hence holds at every $k \in K$. In particular it holds for all positive and negative integer twists. Every individual positive-degree equation is indispensable within the displayed tower with $q_0 = 1$.

Proof. For $n \geq 1$, expanding the product of $n + 1$ copies of Q gives

$$[u^n]Q^{n+1} = (n+1)q_n + P_n(q_1, \dots, q_{n-1}).$$

The only occurrences of q_n choose degree n from one factor and degree zero from the others. As $n + 1$ is invertible, this equation determines q_n from the preceding coefficients. This proves uniqueness, provided existence is established.

Put $z = 1 - e^{-u}$, whose compositional inverse is $u = -\log(1 - z)$. The formal change-of-coordinate rule for residues gives

$$\begin{aligned} [u^n]e^{ku}Q_T^{n+1} &= \text{Res}_u \frac{e^{ku}}{(1 - e^{-u})^{n+1}} du \\ &= \text{Res}_z z^{-n-1}(1 - z)^{-k-1} dz = [z^n](1 - z)^{-k-1} = \binom{n+k}{n}. \end{aligned}$$

For completeness, residue substitution by a coordinate $u = \phi(z) = z + O(z^2)$ follows by linearity from Laurent monomials. If $j \neq -1$, $\phi^j \phi'$ is the derivative of $\phi^{j+1}/(j+1)$ and has zero residue; if $j = -1$, write $\phi = zv$, $v(0) = 1$, to obtain $\phi'/\phi = z^{-1} + v'/v$, with residue one. Only finitely many negative Laurent powers are involved, and nonnegative powers cannot contribute a residue. The final coefficient identity follows from the binomial series, or by solving $(1 - z)F' = (k+1)F$, $F(0) = 1$, coefficient by coefficient. Taking $k = 0$ proves existence and all tower equations.

If the equation at a positive degree N is removed, retain the uniquely determined coefficients below N , choose any q_N different from its Todd value, and apply the same recursion at every $n > N$. The resulting series satisfies every retained equation and differs from Q_T . A finite initial tower leaves every later coefficient unconstrained. Finally, if q_0 is only required to be a unit and the degree-zero equation is absent, the coefficient of q_n becomes $(n+1)q_0^n$. Thus each chosen unit q_0 gives a recursively determined tower; normalization cannot be silently dropped. $\square$

For comparison, arbitrary powers have the separate finite coefficient formula

$$[u^n]Q_T(u)^k = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum j m_j = n}} (k)_M \prod_{j=1}^n \frac{q_j^{m_j}}{m_j!}, \quad M = \sum_{j=1}^n m_j, \quad (52)$$

where $(k)_{\underline{M}} = k(k-1) \cdots (k-M+1)$ and the empty product is one. Indeed, apply the formal binomial expansion to $Q_T = 1 + (Q_T - 1)$ and then the multinomial theorem. This formula does not replace the exponent $n + 1$ and exponential twist in (51).

Theorem 5.2 (Reversible characteristic-series formulas). *Define the normalized series*

$$Q_A(u) = \frac{u/2}{\sinh(u/2)}, \quad Q_L(u) = \frac{u}{\tanh u}, \quad Q_y(u) = Q_T((1+y)u) - yu.$$

Their constant terms are one. With y fixed and marked and $1+y \neq 0$, each of Q_T, Q_A, Q_L, Q_y , and each of their multiplicative inverses, determines the others in the marked formal coordinate. Explicit reconstruction formulas are

$$Q_T = e^{u/2} Q_A, \quad Q_L(u) = Q_T(2u) - u, \quad Q_T(u) = Q_L(u/2) + u/2, \quad (53)$$

$$e^u = \frac{Q_T(u)}{Q_T(u) - u}, \quad e^{2u} = \frac{Q_L(u) + u}{Q_L(u) - u}, \quad (54)$$

$$Q_T(v) = Q_y \left(\frac{v}{1+y} \right) + \frac{yv}{1+y}. \quad (55)$$

For a reconstruction from Q_A alone, set

$$b = \frac{u}{Q_A(u)}, \quad w = \frac{b + \sqrt{b^2 + 4}}{2}, \quad Q_T = wQ_A,$$

where the formal square root has constant term 2. The specialization $y = -1$ is exactly $Q_{-1}(u) = 1 + u$ and has no inverse of the form claimed. Over a general $\mathbb{Q}$ -algebra the hypothesis is that $1+y$ is a unit, and over $\mathbb{Q}[y]$ the inverse formula takes place after localization at $1+y$.

Proof. Substitution of $\sinh v = (e^v - e^{-v})/2$ and $\tanh v = (e^{2v} - 1)/(e^{2v} + 1)$ proves (53)–(54). All denominators in the displayed ratios are units. The square-root recursion is unique because its leading coefficient is 2, so its successive unknown coefficients have invertible coefficient 4. In the present case $b = 2 \sinh(u/2)$ and this root is $2 \cosh(u/2)$; thus $w = e^{u/2}$. Equivalently $w(0) = 1$ and $w - w^{-1} = b$, whose unit solution is unique. The affine rescaling defining Q_y gives (55) directly, and $y = -1$ gives $1 + u$. Multiplicative inversion on units is an involution. These calculations establish the stated reversals on the exact series; applying the square-root algorithm to an arbitrary even unit does not certify that its input equals Q_A . $\square$

Proposition 5.3 (Formal, analytic, and global recovery). *The preceding reversals recover the formal exponential and its inverse formal logarithm, hence the germ of $H(1+z) = \log(1+z) - z$ at $z = 0$. After specializing coefficients and the marked parameter y to the reals, they also construct the displayed standard real analytic functions on their full real domains. They identify an independently supplied global function from its germ only in a declared uniqueness category, for example real analytic functions on the same connected domain. No analogous identification of arbitrary smooth global extensions is valid.*

Proof. Equation (54) recovers e^u , and the series $e^u - 1$ has linear coefficient one, so it has a unique compositional inverse, $\log(1 + z)$. The real quotient definitions extend through $u = 0$ with value one. Their other denominators do not vanish on the real line; in the Q_A reconstruction the chosen root is $2 \cosh(u/2) > 0$. Consequently the explicit formulas define the claimed real analytic extensions. The analytic identity theorem on a connected interval proves uniqueness within that category.

The nearest nonremovable complex poles are $\pm 2\pi i$ for Q_T, Q_A and $\pm \pi i$ for Q_L . Their Taylor discs therefore do not themselves cover the real line. For the smooth counterexample, choose a nonzero smooth bump χ supported in a compact subinterval of $(0, 1)$ away from 1. Then $I + \varepsilon\chi$ has the same germ at 1 as I and differs globally for $\varepsilon \neq 0$. Since $I'' = t^{-2}$ has a positive minimum on the support, a sufficiently small $|\varepsilon|$ preserves strict convexity. A formal identity alone cannot exclude this example. $\square$

Theorem 5.4 (The universal Thom correction, conditional on topology). *Suppose complex topological K -theory and rational ordinary cohomology are supplied with their Thom isomorphisms, compatible complex orientations, natural multiplicative Thom classes, the Chern character normalized by $\text{ch}(L) = e^{c_1(L)}$, and an injective splitting principle. For a complex vector bundle $V \rightarrow B$ on a finite CW complex, with projection π , the uniquely defined correction in*

$$\text{ch}(U_K(V)) = \pi^* C(V) U_H(V) \quad (56)$$

is

$$C(V) = \text{Td}(V)^{-1} = \prod_i Q_T(u_i)^{-1},$$

where the u_i are formal Chern roots. Its complete universal line series determines Q_T and conversely. With compatible collapse and Gysin constructions supplied as well, this yields

$$\text{ch}(f_!^K a) = f_*^H (\text{ch}(a) \text{Td}(T_f)) \quad (57)$$

for $a \in K^0$ and a proper smooth map between manifolds with the specified complex K -orientation and virtual relative tangent bundle T_f , using compact supports when required. On universal bases the characteristic-class statement is taken in completed rational cohomology.

Proof. The cohomological Thom isomorphism makes multiplication by $U_H(V)$ an isomorphism from the cohomology of the base to the appropriate relative group, so it gives existence and uniqueness of $C(V)$ in (56). For a line bundle put $u = c_1(L)$. The zero-section pullbacks of the two Thom classes are respectively $1 - [L^*]$ and u . Thus

$$uC(u) = 1 - e^{-u}.$$

Naturality makes these identities compatible on the finite stages of the universal line bundle. Their inverse limit is $\mathbb{Q}[[u]]$, in which multiplication by u is injective. Hence $C(u) = (1 - e^{-u})/u = Q_T(u)^{-1}$. Multiplicativity of Thom classes and the Chern character gives the product formula after splitting, and injectivity of

the supplied splitting map descends it to B . For an embedding, applying the collapse map gives the cohomological pushforward with the correction $\mathrm{Td}(\nu)^{-1}$ for its complex normal bundle ν . Since $T_f = -\nu$, this is (57); the supplied stable factorization and composition rules give the same formula for the stated proper maps. Finally, the universal line correction is a unit, so inversion recovers Q_T . $\square$

The scalar Euler-coordinate identity used here is itself elementary: for $x \star y = x + y + xy$ and $z(x) = x/(1+x)$,

$$z(x \star y) = z(x) + z(y) - z(x)z(y).$$

Indeed $1 - z(x) = 1/(1+x)$ is multiplicative under $\star$. Interpreting x as $[L] - 1$ and z as $1 - [L^*]$ is a map into a supplied K -theory ring. The scalar identity does not construct bundles, Thom isomorphisms, or the underlying spaces. On a finite base u may be nilpotent and a zero divisor; cancellation in that base would be invalid. If $u^{N+1} = 0$, adding $u^{N+1}R(u)$ to a characteristic series does not change its evaluation there. A single finite-base Thom comparison or numerical index consequently cannot determine the complete universal series. Interpretations of the L , $\hat{A}$, and χ_y series retain their respective oriented, spin, or complex/holomorphic foundations; the algebraic formulas do not assert those foundations or their index theorems.

Proposition 5.5 (Rational characteristic data lose integral data). *On $\mathbb{RP}^2$, the complexification L of the tautological real line is nontrivial and $[L] \neq [1]$ in complex K^0 , although $\mathrm{ch}(L) = 1$ in rational cohomology. Thus even with the base fixed, rational characteristic data and the complete scalar series do not reconstruct integral bundle or K -theory data.*

Proof. The usual cellular calculation gives $H^2(\mathbb{RP}^2; \mathbb{Z}) = \mathbb{Z}/2$ and no positive-degree rational cohomology. To verify nontriviality directly, a nowhere-zero section of L would pull back to an odd map $g : S^2 \rightarrow \mathbb{C} \setminus \{0\}$. Normalize it to an odd map $S^2 \rightarrow S^1$. Since S^2 is simply connected, write $g(x) = e^{i\theta(x)}$ for a continuous real function θ . Then $\theta(-x) - \theta(x)$ is a continuous odd integral multiple of π , hence constant by connectedness. Replacing x by $-x$ changes its sign, which is impossible for a nonzero odd multiple of π . Thus L is nontrivial. Equivalently, under the usual classification of complex line bundles, its first Chern class is the nonzero element of the displayed group.

If $[L] = [1]$, the Grothendieck-group relation supplies a bundle E with $L \oplus E \cong 1 \oplus E$. Taking determinant lines and cancelling $\det E$ would make L trivial, a contradiction. On the other hand, every positive-degree rational Chern class vanishes, so $\mathrm{ch}(L) = 1$. The transition functions of L are signs, so $L \otimes L \cong 1$, as expected from the torsion line class. These arguments use the stated elementary topological foundations and do not follow from formal scalar arithmetic alone. $\square$

6 The matrix lift and its Gaussian statistics

Write $\mathcal{P}_n$ for the cone of real symmetric positive-definite $n \times n$ matrices, E_n for its identity, and use the trace pairing $\langle U, V \rangle = \text{tr}(UV)$ on symmetric matrices. All matrix logarithms, square roots, and powers below are the positive spectral ones. The finite-dimensional real spectral theorem is retained throughout this section.

Theorem 6.1 (The unique scalar-block spectral lift). *Suppose a family $F_n : \mathcal{P}_n \rightarrow \mathbb{R}$, $n \geq 1$, satisfies*

$$F_1([t]) = I(t), \quad F_n(QXQ^\top) = F_n(X)$$

for orthogonal Q , and the scalar-block recursion

$$F_{n+1}(X \oplus [t]) = F_n(X) + I(t).$$

Then, without any regularity assumption,

$$F_n(X) = \Phi_n(X) := \sum_{i=1}^n I(\lambda_i(X)) = \text{tr } X - \log \det X - n. \quad (58)$$

This family satisfies full block additivity. Neither orthogonal invariance nor scalar-block recursion can be removed while keeping the scalar seed, even among nonnegative candidates.

Proof. Orthogonally diagonalize X and apply scalar-block recursion until all blocks have size one. This forces the sum in (58). Trace, determinant, and their behavior on block diagonal matrices verify every axiom for Φ_n .

If orthogonal invariance is omitted, $F_n(X) = \sum_i I(X_{ii})$ retains the seed and full block additivity. Rotating $\text{diag}(2, 1)$ through angle $\pi/4$ gives diagonal entries $3/2, 3/2$, and $I(2) - 2I(3/2) = \log(9/8) \neq 0$, so it is a different lift. If scalar-block recursion is omitted, for $\varepsilon > 0$ put

$$G_n(X) = \Phi_n(X) + \varepsilon \sum_{i < j} (\log \lambda_i(X) - \log \lambda_j(X))^2.$$

This has the seed, orthogonal invariance, and nonnegativity, but on $[a] \oplus [b]$, $a \neq b$, its extra term is nonzero whereas both rank-one extra terms vanish. If the seed is omitted, the zero family already retains the two structural rules. These are omission tests in the stated family, rather than a claim of an absolute weakest possible axiomatization. $\square$

Theorem 6.2 (Derivatives, divergence, duality, and determinant bounds). *In the indicated affine coordinates the potential Φ_n satisfies*

$$\nabla \Phi_n(X) = E_n - X^{-1}, \quad (59)$$

$$g_X(U, V) := D^2 \Phi_n(X)[U, V] = \text{tr}(X^{-1}UX^{-1}V), \quad (60)$$

$$D(X, Y) := \Phi_n(X) - \Phi_n(Y) - \langle \nabla \Phi_n(Y), X - Y \rangle = \text{tr}(Y^{-1}X) - \log \det(Y^{-1}X) - n. \quad (61)$$

The Hessian is positive definite and $D(X, Y) \geq 0$, with equality exactly when $X = Y$. For every invertible real C , simultaneous congruence preserves both D and g . Inversion is an isometry for g and reverses the directed divergence:

$$D(X^{-1}, Y^{-1}) = D(Y, X).$$

Extending Φ_n by $+\infty$ outside $\mathcal{P}_n$ in the vector space of symmetric matrices, its conjugate is

$$\Phi_n^*(\Theta) = \begin{cases} -\log \det(E_n - \Theta), & E_n - \Theta \in \mathcal{P}_n, \\ +\infty, & \text{otherwise.} \end{cases} \quad (62)$$

The Legendre coordinate is $\Theta = E_n - X^{-1}$, with inverse $X = (E_n - \Theta)^{-1}$. Moreover

$$\det X \leq e^{\operatorname{tr} X - n}, \quad \det X \leq \left(\frac{\operatorname{tr} X}{n} \right)^n,$$

with equality respectively at E_n and at positive scalar matrices; $\log \det$ is strictly concave.

Proof. Jacobi's formula gives $D \log \det X[U] = \operatorname{tr}(X^{-1}U)$. Differentiating $XX^{-1} = E_n$ gives $D(X^{-1})[U] = -X^{-1}UX^{-1}$ and proves the first two formulas. Their positivity is the precise identity

$$g_X(U, U) = \operatorname{tr}((X^{-1/2}UX^{-1/2})^2) = \|X^{-1/2}UX^{-1/2}\|_F^2.$$

It is the trace of a square, and is strictly positive for $U \neq 0$. Substitution yields (61). With $A = Y^{-1/2}XY^{-1/2}$, trace and determinant show $D(X, Y) = \sum_i I(\lambda_i(A))$. Scalar nonnegativity and its equality case prove the assertion, since $A = E_n$ precisely when $X = Y$. Cyclic trace and the inverse-congruence formula prove congruence invariance. Substituting inverses in D gives its reversal; inserting the differential $-X^{-1}UX^{-1}$ in $g_{X^{-1}}$ gives g_X .

If $B = E_n - \Theta$ is positive definite, maximizing the conjugate is equivalent to maximizing $-\operatorname{tr}(BX) + \log \det X + n$. Substitute $A = B^{1/2}XB^{1/2}$ to obtain $-\Phi_n(A) - \log \det B$. Its unique maximum occurs at $A = E_n$, which proves the finite formula and its optimizer. If B has an eigenvalue $b \leq 0$, take X diagonal in an eigenbasis of B , with the matching eigenvalue equal to $r \rightarrow \infty$ and all others one. The contribution $-br + \log r$ diverges even when $b = 0$. Substituting Fenchel equality consequently gives the Bregman duality $D_{\Phi_n}(X, Y) = D_{\Phi_n^*}(\Theta_Y, \Theta_X)$.

Finally $\Phi_n(X) \geq 0$ gives the first determinant bound. Apply it to $nX / \operatorname{tr} X$ for the second and its equality case. The Hessian of $\log \det$ is $-g$, which proves strict concavity on the convex cone. $\square$

The gradient in (59), together with $F(E_n) = 0$, identifies a differentiable F on the cone: the gradient of $F - \Phi_n$ vanishes, so integration on each straight segment makes the difference constant. Likewise the full Hessian (60), together with $F(E_n) = 0$ and $\nabla F(E_n) = 0$, identifies a twice differentiable F , since the Hessian of the difference vanishes and its gradient is constant on segments. These

affine marks remove exactly the affine ambiguity. The rank-one potential or its calibrated anchored Bregman function recovers I directly. A metric given merely up to isometry does not provide these affine coordinates or marks. Nor does congruence invariance alone select this metric: the forms

$$\alpha g_X(U, V) + \beta \operatorname{tr}(X^{-1}U) \operatorname{tr}(X^{-1}V), \quad \alpha > 0, \quad \beta > -\alpha/n,$$

are also invariant positive metrics. Positivity follows by separating $X^{-1/2}UX^{-1/2}$ into its scalar and traceless parts. The selected Hessian, rather than bare invariance or determinant inequalities, removes this freedom.

Theorem 6.3 (Geodesics, distance, and the symmetrization boundary). *For $X, Y \in \mathcal{P}_n$ put $K = \log(X^{-1/2}YX^{-1/2})$. The unique constant-speed minimizing geodesic for g is*

$$\gamma(s) = X^{1/2}e^{sK}X^{1/2}, \quad 0 \leq s \leq 1, \quad d(X, Y) = \|K\|_F.$$

If λ_i are the eigenvalues of $X^{-1/2}YX^{-1/2}$, then

$$D(X, Y) + D(Y, X) = 4 \sum_i \sinh^2 \left(\frac{\log \lambda_i}{2} \right).$$

For $n = 1$ this is $4 \sinh^2(d(X, Y)/2)$. For every $n \geq 2$, symmetrization is not a function of distance alone.

Proof. Congruence reduces the distance problem to a curve from E_n to $A = X^{-1/2}YX^{-1/2}$. Write any piecewise continuously differentiable such curve as $B(s) = e^{Z(s)}$ with $Z(s)$ symmetric. The exponential differential, obtained by differentiating its power series, is

$$D \exp_Z[V] = \int_0^1 e^{(1-r)Z} V e^{rZ} dr.$$

In a basis diagonalizing Z with eigenvalues z_i , its (i, j) entry is $V_{ij}(e^{z_i} - e^{z_j})/(z_i - z_j)$, with the continuous value $e^{z_i}V_{ij}$ when $z_i = z_j$. Consequently

$$g_B(B', B') = \sum_{i,j} \left(\frac{2 \sinh((z_i - z_j)/2)}{z_i - z_j} \right)^2 (Z'_{ij})^2 \geq \|Z'\|_F^2,$$

where the quotient is one at zero. Its absolute value is at least one by the power series for $\sinh$. Integrating speed gives

$$\operatorname{length}_g(B) \geq \int_0^1 \|Z'(s)\|_F ds \geq \|\log A\|_F.$$

The curve $e^{s \log A}$ attains both inequalities and has constant speed. For a minimizing curve equality in the Euclidean length inequality forces $Z(s) = a(s) \log A$ with a nondecreasing from zero to one. On this curve metric speed is

$a'(s)\|\log A\|_F$; constant speed forces $a(s) = s$. If $A = E_n$, zero length forces the constant curve. This proves uniqueness as stated. The minimizing curve is a geodesic by the usual first-variation theorem; directly it satisfies $\gamma'' = \gamma'\gamma^{-1}\gamma'$, the Euler–Lagrange equation obtained from the energy $\frac{1}{2} \int \text{tr}(\gamma^{-1}\gamma'\gamma^{-1}\gamma') ds$.

Adding the two divergence formulas cancels logarithms and gives $\sum_i (\lambda_i + \lambda_i^{-1} - 2)$, proving symmetrization. For $r > 0$ compare $X = E_n$ with matrices whose log eigenvalue vectors are $(r, 0, \dots, 0)$ and $(r/\sqrt{2}, r/\sqrt{2}, 0, \dots, 0)$. Both distances equal r , whereas their symmetrizations are $2 \cosh r - 2$ and $4 \cosh(r/\sqrt{2}) - 4$. The difference has power series

$$\sum_{j \geq 2} \frac{2 - 2^{2-j}}{(2j)!} r^{2j} > 0.$$

This proves the failure for every positive distance in rank at least two. $\square$

Theorem 6.4 (The supplied Gaussian and Wishart sampling model). *Let $m \geq 1$ and $n \geq 1$ be specified integers and suppose $Z_1, \dots, Z_m$ are independent, identically distributed $N(0, X)$ vectors with $X \in \mathcal{P}_n$ and known mean zero. Put $W = \sum_{j=1}^m Z_j Z_j^T$ and $C = W/m$. The scatter W has the Wishart law defined by this model and is a sufficient statistic for X , with*

$$\log L(X) = -\frac{m}{2} \log \det X - \frac{1}{2} \text{tr}(X^{-1}W) + \text{constant}.$$

For symmetric Θ ,

$$\mathbb{E} e^{\text{tr}(\Theta W)} = \det(E_n - 2X^{1/2}\Theta X^{1/2})^{-m/2}$$

when the matrix inside the determinant is positive definite; the expectation is infinite otherwise. The scatter is positive definite almost surely exactly when $m \geq n$. Whenever the observed C is positive definite, the maximum likelihood covariance is uniquely C and, with $\ell = -\log L$,

$$\ell(X) - \ell(C) = \frac{m}{2} D(C, X).$$

If C is singular there is no maximum likelihood estimate in the open cone $\mathcal{P}_n$. Finally

$$\text{KL}(N(0, X) \| N(0, Y)) = \frac{1}{2} D(X, Y),$$

and the divergence for m independent copies is m times this value.

Proof. Multiplying the Gaussian densities proves the likelihood formula. Its dependence on the samples factors through W , so the factorization criterion for dominated statistical families proves sufficiency. For one sample write $Z = X^{1/2}G$ with $G \sim N(0, E_n)$. The tilted Gaussian integral has quadratic form $E_n - 2X^{1/2}\Theta X^{1/2}$. Orthogonal diagonalization reduces it to one-dimensional integrals, giving its inverse square-root determinant when all eigenvalues are positive. A zero or negative eigenvalue yields an infinite one-dimensional integral,

so Tonelli's theorem gives divergence otherwise. Independence raises the finite answer to the m th power.

The rank of W is the dimension of the span of its m sample vectors. It is at most m , and the first n Gaussian sample vectors are linearly independent almost surely: the determinant-zero set has Lebesgue measure zero, as follows, for example, by conditioning successively on the span of the preceding vectors. Thus W is positive definite almost surely for $m \geq n$, and singular for $m < n$.

For positive-definite C , subtracting likelihood values gives $mD(C, X)/2$, with the asserted minimum and equality case. For singular C , diagonalize it and let the eigenvalues of X along the nontrivial kernel of C tend to zero while keeping the remaining eigenvalues fixed and positive. The trace term stays fixed and $\log \det X \rightarrow -\infty$, so $\ell(X) \rightarrow -\infty$ and no minimizer exists in the open cone. Finally integrating the Gaussian log density ratio uses $\mathbb{E}_X ZZ^\top = X$ and gives $D(X, Y)/2$; the log ratio adds over independent samples. $\square$

The sample count, independence, Gaussian law, and known-mean convention are data of this theorem. The scalar potential alone supplies none of those sampling choices. Likewise the spectral or Gaussian construction of Φ_n identifies that constructed family; an independently designated family is identified only after the rules of Theorem 6.1 are imposed.

7 Gaussian radial realization and independent spatial data

Theorem 7.1 (The isotropic four-dimensional Gaussian realization). *If $Z \sim N(0, E_4)$ and $T = \|Z\|^2/2$, then T has density $p(t) = te^{-t}$ on $t > 0$. Conversely, if T has this density and Ω is an independent uniform point of S^3 , then $Z = \sqrt{2T}\Omega$ has law $N(0, E_4)$. Equivalently the radial law together with orthogonal invariance uniquely identifies this Gaussian law; the radial law alone does not.*

For the specified Ornstein–Uhlenbeck process in $\mathbb{R}^D$,

$$dZ_s = -\frac{1}{2}Z_s ds + dB_s,$$

the generator on radial-energy functions $F(z) = f(\|z\|^2/2)$ is

$$\left(\frac{1}{2}\Delta - \frac{1}{2}z \cdot \nabla\right) F = tf''(t) + (D/2 - t)f'(t).$$

Thus its radial generator is $tf'' + (2 - t)f'$ exactly when $D = 4$. In dimension four the radial energy satisfies

$$dT_s = (2 - T_s)ds + \sqrt{2T_s}d\beta_s$$

for a one-dimensional Brownian motion β . These statements select a dimension within this specified Gaussian construction.

Proof. Polar integration in $\mathbb{R}^4$, using $|S^3| = 2\pi^2$, gives

$$(2\pi)^{-2}e^{-r^2/2}2\pi^2r^3dr = te^{-t}dt, \quad t = r^2/2.$$

This proves the forward law. Conversely, integration of any bounded Borel test function first over uniform Ω and then over this radial density reproduces the four-dimensional Gaussian integral. For an orthogonally invariant law, average the test function over the normalized Haar measure on the compact group $O(4)$. On each nonzero sphere this average is its uniform spherical average, so its integral is fixed by the radial law. This proves the equivalent uniqueness statement and the independence of radius and uniform angle. Choosing instead a fixed unit vector v and $Z = \sqrt{2T} v$ preserves the radial law and changes the vector law.

The chain rule gives $\nabla F = f'(t)z$, $\Delta F = 2tf''(t) + Df'(t)$, and $z \cdot \nabla F = 2tf'(t)$, proving the generator formula. Itô's formula yields

$$dT_s = (D/2 - T_s) ds + Z_s \cdot dB_s, \quad d\langle T \rangle_s = 2T_s ds.$$

Normalize the stochastic integrand to $Z_s/\|Z_s\|$ off zero and choose a fixed unit vector at zero. Its stochastic integral β_s has quadratic variation s , hence is Brownian by the continuous-martingale characterization, and $Z_s \cdot dB_s = \sqrt{2T_s} d\beta_s$. At $D = 4$ this gives the displayed equation. The stationary vector law is $N(0, E_D)$ by the explicit Gaussian solution of the OU equation; polar integration therefore gives the stationary energy law $t^{D/2-1}e^{-t}/\Gamma(D/2)$. Its shape is two precisely at $D = 4$. $\square$

This construction retains the Gaussian and stochastic-calculus foundations just used. A stationary density by itself does not choose a diffusion: for any specified positive differentiable coefficient $a(t)$ the formal zero-current expression

$$L_a f = \frac{1}{p}(apf')' = af'' + \left(a' + a\frac{p'}{p}\right)f'$$

has the same formal symmetric measure, as integration by parts on compactly supported tests verifies. Domain and endpoint conditions remain necessary when selecting an actual closed generator. Choosing $a(t) = t$ gives the radial operator in Theorem 7.1; it is a selection, not a consequence of knowing only p .

Theorem 7.2 (Nonnegative orthogonal additivity without regularity). *Let V be a real inner-product space of dimension at least two and let $r : V \rightarrow [0, \infty)$. Then*

$$r(x + y) = r(x) + r(y) \quad \text{whenever } \langle x, y \rangle = 0$$

holds if and only if $r(x) = c\|x\|^2$ for some $c \geq 0$. No continuity, measurability, homogeneity, or radiality hypothesis is required. Consequently a function $z : V \rightarrow [0, \infty)$ satisfies

$$z(x + y) = z(x) + z(y) + z(x)z(y) \quad (x \perp y)$$

if and only if $z(x) = e^{c\|x\|^2} - 1$ for some $c \geq 0$. Nontriviality forces $c > 0$, while a numerical value at one nonzero vector is needed to calibrate its scale.

Proof. The equation at $(0, 0)$ gives $r(0) = 0$. Define the even and odd parts

$$e(x) = \frac{r(x) + r(-x)}{2}, \quad o(x) = \frac{r(x) - r(-x)}{2}.$$

They are orthogonally additive, $e \geq 0$, and $|o(x)| \leq e(x)$. If $\|x\| = \|y\|$, set $a = (x+y)/2$, $b = (x-y)/2$. Then $a \perp b$, $x = a+b$, and $y = a-b$. Evenness gives $e(x) = e(a) + e(b) = e(a) + e(-b) = e(y)$. Every nonnegative real number is the squared norm of a vector, so $e(x) = A(\|x\|^2)$ for a function $A : [0, \infty) \rightarrow [0, \infty)$.

There exist two orthogonal unit vectors. Multiplying them by $\sqrt{a}$ and $\sqrt{b}$ shows $A(a+b) = A(a) + A(b)$ for arbitrary $a, b \geq 0$. Nonnegativity makes A monotone: if $b \geq a$ then $A(b) - A(a) = A(b-a) \geq 0$. With $c = A(1)$, additivity gives $A(q) = cq$ for every nonnegative rational q . For any real $a \geq 0$, choose rational sequences approaching a from below and above. Monotonicity sandwiches $A(a)$ between their multiples by c , so $A(a) = ca$.

For $x \neq 0$, choose $y \perp x$ of the same norm; the case $x = 0$ is automatic. Then $x + y \perp x - y$, and orthogonal additivity and oddness give

$$o(2x) = o(x+y) + o(x-y) = o(x) + o(y) + o(x) + o(-y) = 2o(x).$$

Iterating downward, $o(x) = 2^k o(x/2^k)$, while $|o(x/2^k)| \leq c\|x\|^2/4^k$. Hence $|o(x)| \leq c\|x\|^2/2^k$ for every k , forcing $o(x) = 0$. This proves $r = e = c\|x\|^2$. Pythagoras proves the converse.

For the second statement take $r = \log(1+z)$. Its values are nonnegative, and the composition equation is exactly orthogonal additivity of r . Apply the first statement and exponentiate; the converse follows by multiplication of exponentials. $\square$

Both substantive hypotheses have direct omission tests. In dimension one, $r(x) = x^4$ is nonnegative and orthogonally additive, since every orthogonal pair includes zero, but is not $c\|x\|^2$. Without nonnegativity in dimension at least two, a nonzero linear functional, or $\|x\|^2$ plus such a functional, is orthogonally additive and violates the conclusion. Accordingly the condition $z > -1$ alone also fails: $z(x) = e^{\ell(x)} - 1$ for a nonzero linear functional obeys the composition rule and takes negative values.

Proposition 7.3 (The profile-independent radial residual). *In integer Euclidean dimension $D \geq 1$ the radial Laplacian is $\Delta_D f = f'' + (D-1)f'/x$ for $x > 0$. For every twice differentiable nonvanishing profile f and $\alpha = (D-1)/2$,*

$$\frac{(x^\alpha f)''}{x^\alpha f} - \left(\frac{f''}{f} + \frac{D-1}{x} \frac{f'}{f} \right) = \frac{(D-1)(D-3)}{4x^2}. \quad (63)$$

Equivalently, writing $f = x^{-\alpha}u$ gives the operator identity

$$-\Delta_D f = x^{-\alpha} \left(-u'' + \frac{(D-1)(D-3)}{4x^2} u \right),$$

which does not require f to be nonvanishing. Residual cancellation holds exactly for $D = 1$ or $D = 3$; with the independently retained restriction $D \geq 2$ it selects $D = 3$. It has no reverse identifying force for the profile f .

Proof. Twice applying the product rule gives

$$\frac{(x^\alpha f)''}{x^\alpha f} = \frac{f''}{f} + \frac{2\alpha}{x} \frac{f'}{f} + \frac{\alpha(\alpha-1)}{x^2}.$$

Since $2\alpha = D-1$ and $\alpha(\alpha-1) = (D-1)(D-3)/4$, subtraction proves (63); the same product rule proves the operator form. The coefficient vanishes precisely at the two stated integer dimensions. It is independent of f , so for example $f(x) = e^{-x^2}$ also gives cancellation in dimension three, despite being a different profile from p or the scalar potential. $\square$

Proposition 7.4 (Spatial countermodels retaining the complete intrinsic structure). *All intrinsic scalar identities, their probability and operator realizations, their formal characteristic series, and the canonical four-dimensional Gaussian construction may hold while an independent designated spatial observable fails orthogonal additivity, or while its designated dimension fails to equal three. Their conjunction therefore does not imply either the hypotheses of Theorem 7.2 or a designated dimension conclusion from Proposition 7.3.*

Proof. Keep the entire intrinsic structure and every canonical auxiliary object fixed. Adjoin a distinct Euclidean spatial sort V with its own designated observable r . Three choices give the required models. In $V = \mathbb{R}^2$ take $r(x) = \|x\|^4$. For perpendicular unit vectors x, y , $r(x+y) = 4$ while $r(x) + r(y) = 2$, so even positivity, smoothness, and radiality do not force the composition law. In $V = \mathbb{R}^2$ take $r(x) = \|x\|^2$. Orthogonal additivity holds, but the coefficient in (63) is $-1/(4x^2)$. Finally in $V = \mathbb{R}$ take squared norm: residual cancellation holds although $D \neq 3$. None of these additions changes any object or equation in the original intrinsic structure. In particular, each coexists with the separately constructed Gaussian vector in $\mathbb{R}^4$; no identification between its vector space and the designated spatial sort was supplied. $\square$

There is also a direct tensor distinction. Pulling a one-dimensional metric $g(r) dr^2$ back by a spatial scalar map produces $g(r) dr \otimes dr$, of rank at most one. The Hessian of a composite potential is instead

$$D^2(I \circ r) = I''(r) dr \otimes dr + I'(r) D^2 r.$$

The chain rule proves this formula. Its second term prevents an identification of these two tensors or an inference of ambient dimension from the rank of the first.

8 Marked rational dynamics and transported integer arithmetic

Theorem 8.1 (Rational trees, Euclidean decoding, and the matrix monoid). *Fix the positive rational coordinate, the root 1, ordered edge labels L, R , and the transformations*

$$L(x) = \frac{x}{1+x}, \quad R(x) = 1+x, \quad M_L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad M_R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Successively applying the letters of a word to 1 gives a rooted Calkin–Wilf tree [3] in which every positive rational occurs exactly once. The matrices M_L, M_R freely generate exactly the determinant-one matrices with nonnegative integer entries, including the identity.

For a Stern–Brocot interval [4] store the upper endpoint as the first column and the lower endpoint as the second, starting with columns $(1, 0)^\top$ and $(0, 1)^\top$, representing ∞ and 0. Its node is their mediant and a left or right step multiplies this matrix on the right by M_L or M_R . With these conventions the Stern–Brocot label of a word is the Calkin–Wilf label of the reversed word.

The complete root path, its oriented subtraction runs, and the finite continued fraction are mutually decodable, with the canonical convention

$$[a_0; a_1, \dots, a_k], \quad a_0 \geq 0, \quad a_j \geq 1 \ (j \geq 1), \quad a_k \geq 2 \text{ if } k \geq 1.$$

A one-term expansion $[a_0]$ has $a_0 \geq 1$, so the root is [1]. The last run in the subtraction-to-root algorithm has length one less than its terminal Euclidean quotient; it has length zero for the root.

Proof. For a reduced fraction $a/b < 1$, the unique possible parent is $a/(b-a)$ and the last edge is L ; if $a/b > 1$, it is $(a-b)/b$ with last edge R . Both numerators and denominators remain positive, the gcd is unchanged, and their sum strictly decreases. The algorithm terminates only at $a = b$, hence at $1/1$ by coprimality. It therefore constructs a root path for every positive rational. At each step both the last edge and parent are forced, proving uniqueness. Conversely the child formulas preserve positive coprimality, so every word has a positive reduced rational label.

For matrix generation write a nonnegative determinant-one matrix as $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Except for the identity, its two rows are comparable coordinatewise. Indeed one pattern of incomparable rows, $a < c$ and $b > d$, would give $ad < bc$. In the other pattern write $a = c + u$, $d = b + v$ with $u, v \geq 1$. The determinant equation becomes

$$1 = cv + ub + uv,$$

forcing $u = v = 1$ and $b = c = 0$, which is the identity. If the lower row dominates, subtract the upper row to obtain $M = M_L N$; if the upper dominates, subtract the lower to obtain $M = M_R N$. The resulting N remains nonnegative integral with determinant one and strictly smaller entry sum. Equal rows are impossible. Repeated subtraction therefore ends at the identity and proves generation. The dominance choice at every nonidentity step is unique; cancellation then proves uniqueness of the generator word. Conversely products have nonnegative integral entries and determinant one. The word “nonnegative” is essential: the generators have zero entries.

If $w = s_1 \cdots s_\ell$ records chronological path steps, its Calkin–Wilf column vector is $M_{s_\ell} \cdots M_{s_1} (1, 1)^\top$. The Stern–Brocot endpoint matrix is $M_{s_1} \cdots M_{s_\ell}$ and its mediant vector is that matrix times $(1, 1)^\top$. This proves reversal and also proves complete Stern–Brocot enumeration. For example, $CW(LR) = 3/2$ and $SB(LR) = 2/3$. Right multiplication by M_L replaces the upper endpoint

column by its sum with the lower, and right multiplication by M_R replaces the lower by that sum. Determinant one ensures ordered adjacent endpoints and a reduced mediant throughout.

Lastly, consecutive subtractions of the same smaller integer from the larger group into Euclidean integer division. If $a = qb + r$ with $0 < r < b$, exactly q subtractions replace (a, b) by (r, b) . If $r = 0$, coprimality makes $b = 1$ and the algorithm stops after $q - 1$ subtractions at $(1, 1)$; the case with $b > a$ is identical with the roles reversed. Thus oriented runs determine every quotient, with precisely the stated terminal correction. The Euclidean algorithm terminates and its remainders are unique. Successive identities $a/b = q + r/b$ yield the continued fraction. If its last digit is one, the identity $[\dots, c, 1] = [\dots, c + 1]$ removes it; the unique division quotients give the stated canonical terminal convention. A rational value reconstructed from that fraction gives its unique root path by the already proved parent algorithm, which proves mutual decoding. $\square$

Proposition 8.2 (Marked Farey branches and full-domain scalar recovery). *On $[0, 1]$ define the Farey map and its inverse branches [5]*

$$\psi_0(x) = \frac{x}{1+x}, \quad \psi_1(x) = \frac{1}{1+x}.$$

Their respective ranges are $[0, 1/2]$ and $[1/2, 1]$, and they are the increasing and decreasing inverse branches of

$$F(y) = \begin{cases} y/(1-y), & 0 \leq y \leq 1/2, \\ (1-y)/y, & 1/2 \leq y \leq 1. \end{cases}$$

At the common endpoint both forward formulas equal one, and branch labels retain the inverse-branch information.

The complete numerically labelled Calkin–Wilf tree determines L, R on positive rationals. If the candidate maps are declared to be Möbius transformations, these data determine their full rational formulas, in particular L on $(-1, \infty)$. In that marked extension category the scalar potential is recovered by

$$I(t) = \int_0^{t-1} L(s) ds = t - 1 - \log t, \quad t > 0. \quad (64)$$

Connected real analytic continuation is another sufficient extension category. A bare tree, or positive-rational action without a suitable extension category, has no such global converse.

Proof. Derivatives, endpoint values, and substitution verify the ranges, monotonicity, and the two inverse formulas. Notice that ψ_1 is not the Calkin–Wilf map R . Every positive rational occurs in the tree, so its numerical children give both maps on all such inputs. Two Möbius transformations agreeing at three distinct finite inputs where they are defined are equal: cross multiplication gives a polynomial of degree at most two with three roots, hence the zero polynomial. Their fractional-linear maps therefore agree globally away from their pole. This

supplies the extension $s/(1+s)$ for $-1 < s < 0$ as well as for positive s , and direct integration proves (64). Within the real analytic category on $(-1, \infty)$, equality on positive rationals first gives equality on $(0, \infty)$ by continuity and then everywhere by the identity theorem.

Without that category, choose a nonzero smooth bump χ supported in $(1/4, 1/2)$ and put $\tilde{I} = I + \varepsilon\chi$. The induced generator $\tilde{L}(s) = \tilde{I}'(1+s)$ agrees with $L(s)$ for every $s \geq 0$, and hence on all positive-rational inputs, but the potential differs on $(0, 1)$. Small nonzero ε even preserves strict convexity, as in Proposition 5.3. Finally an abstract unlabelled binary tree contains no numerical input/output pairs at all, so it cannot supply the coordinate markings used in this proof. $\square$

The scalar formal-group inverse itself constructs the canonical global rational expression: $\text{inv}_*(x) = -x/(1+x)$, so $L = -\text{inv}_*$ and $R(x) = 1+x$ is the marked shift. This constructs a labelled rational model. It does not attach those labels to an independently supplied abstract tree.

Theorem 8.3 (Exact collisions and complete labelled arithmetic transport). *Fix $\mu, \gamma > 0$ and a real constant c_P , and consider the placed integer code*

$$e(n) = \log(1 + \gamma n) - \mu n + c_P, \quad n \geq 2.$$

For distinct $n > m \geq 2$,

$$e(n) = e(m) \iff \mu = \frac{1}{n-m} \log \frac{1 + \gamma n}{1 + \gamma m}. \quad (65)$$

The rule $e(a) \odot e(b) = e(ab)$ defines a well-defined binary operation on the image if and only if e is injective. Under that hypothesis, write S for the image with an adjoined abstract unit 1_S . Extending $e(1) = 1_S$ makes $e : \mathbb{N}_{>0} \rightarrow S$ an isomorphism of labelled multiplicative monoids. It transports the full prime, valuation, divisor, gcd/lcm, Möbius, and Dirichlet-convolution inventory stated below.

The different intrinsic code

$$e_0(n) = H(n+1), \quad n \geq 1,$$

is injective for every placement and already includes the unit label. It does not remove any collision of the original code e .

Proof. Subtracting the two values of e proves (65); thus avoiding every displayed pairwise parameter curve is exactly injectivity on the specified carrier. If injective, e is a bijection onto its image and ordinary multiplication transports through that bijection, proving well-definedness.

For the reverse implication suppose $n > m$ and $e(n) = e(m)$. Then

$$\mu(n-m) = \int_m^n \frac{\gamma}{1+\gamma u} du.$$

For every real $k > 1$, the same difference calculation and substitution in the integral give

$$e(kn) - e(km) = k \int_m^n \left(\frac{\gamma}{1 + k\gamma u} - \frac{\gamma}{1 + \gamma u} \right) du < 0.$$

Here the formula for e is evaluated on the positive real ray in this calculation. The strict inequality follows pointwise from $u > 0$, $\gamma > 0$, and $k > 1$. Taking an integer $k \geq 2$ shows that multiplication of the common image $e(n) = e(m)$ by $e(k)$ would require two distinct outputs. Hence no operation with the prescribed rule exists. This proves necessity for every collision, not only for a selected example. For instance, $\gamma = 1$ and $\mu = \log(5/4)$ give $e(3) = e(4)$ but $e(6) - e(8) = \log(175/144) > 0$.

When injectivity holds, adjoin an element 1_S distinct from every point of the original image and set $1_S \odot x = x \odot 1_S = x$. The extended map is a bijection preserving multiplication and unit. The integer fundamental theorem of arithmetic consequently gives a unique finite factorization of each $x \in S$ into irreducibles, with the empty product for the unit. All the inventory formulas below follow from that factorization, as verified after the table. Finally $H'(t) = 1/t - 1 < 0$ for $t > 1$, so $H(n+1)$ is strictly decreasing on positive integers. This proves the alternative's injectivity. For the placed profile with arbitrary additive level, write $\tilde{\Sigma}_P(r) = \log(1 + \gamma r) - \mu r + c_P$; its parameters here are $\mu, \gamma > 0$ and $c_P \in \mathbb{R}$. The critical point is $r_* = 1/\mu - 1/\gamma$ and

$$\tilde{\Sigma}_P(r_* + n/\mu) = H(n+1) + \log(\gamma/\mu) + \mu/\gamma - 1 + c_P.$$

Indeed its intrinsic coordinate $\mu(r + 1/\gamma)$ is $n+1$ there. This is a shifted sampling code, with a fixed additive level, rather than the original sampling at $r = n$. $\square$

Write $x \mid y$ for divisibility in the unit-adjoined monoid, let q range over its irreducibles, and write μ_S for the arithmetic Möbius function, to distinguish it from the placement parameter. The complete inventory is

Irreducibles	$q = e(p), \quad p \text{ a numerical prime};$
Valuations	$v_q(x) = \max\{j \geq 0 : q^j \mid x\};$
Factor counts	$\Omega(x) = \sum_q v_q(x), \quad \omega(x) = \#\{q : v_q(x) > 0\};$
Greatest common divisor	$v_q(\gcd(x, y)) = \min(v_q(x), v_q(y));$
Least common multiple	$v_q(\text{lcm}(x, y)) = \max(v_q(x), v_q(y));$
Divisor lattice	$\{d : d \mid x\} \cong \prod_{q \mid x} \{0, \dots, v_q(x)\};$
Divisor count	$\tau(x) = \prod_{q \mid x} (v_q(x) + 1);$
Möbius function	$\mu_S(x) = 0 \text{ if some } v_q(x) \geq 2, \quad \mu_S(x) = (-1)^{\omega(x)} \text{ otherwise};$
Dirichlet convolution	$(f * g)(x) = \sum_{d \mid x} f(d)g(x/d).$

All sums and products over the support of a single element are finite, and $\mu_S(1_S) = 1$. To verify the inventory, divisibility of $x = \prod_q q^{v_q(x)}$ by d is precisely the componentwise condition $0 \leq v_q(d) \leq v_q(x)$. This proves the valuation

maximum, gcd/lcm formulas, lattice product, and divisor count. Irreducibility means exactly that the exponent vector has one coordinate equal to one and all others zero; it therefore identifies the numerical primes under e . Summing μ_S over divisors leaves only the square-free choices, giving

$$\sum_{d|x} \mu_S(d) = (1-1)^{\omega(x)} = \begin{cases} 1, & x = 1_S, \\ 0, & x \neq 1_S. \end{cases}$$

In any fixed commutative coefficient ring, finite reindexing of divisor pairs makes convolution associative and commutative with identity δ_{1_S} . The last displayed identity says $\mu_S * \mathbf{1} = \delta_{1_S}$. Thus if $F(x) = \sum_{d|x} f(d)$, convolution with μ_S gives

$$f(x) = \sum_{d|x} \mu_S(d) F(x/d),$$

proving Möbius inversion. A normalized multiplicative function satisfies $f(1_S) = 1$ and $f(xy) = f(x)f(y)$ for coprime x, y . It is exactly specified by arbitrary values at the marked prime powers, followed by $f(x) = \prod_{q|x} f(q^{v_q(x)})$. The operation does not choose those values. Over a field, every nonzero function satisfying the unnormalized multiplicativity equation automatically has unit value one; the identically zero function is a separate convention.

On the original image omitting one, reflexive divisibility means $x = y$ or $y = x \odot z$ with z in that image. Coprime gcds and the bottom of a full divisor lattice require the adjoined 1_S . It cannot be identified with the real number $\tilde{\Sigma}_P(1)$ unless injectivity has also been checked after adding that index. Prime permutations preserve the abstract free commutative monoid, so numerical prime names and addition are extra labels. With the integer labels supplied, the totient is also transported by

$$\varphi(e(n)) = \prod_{p^a \parallel n} p^{a-1}(p-1), \quad \varphi(1_S) = 1.$$

The formula counts residues not divisible by any prime divisor by inclusion–exclusion: for each square-free divisor d of n there are n/d multiples of d among $1, \dots, n$, giving $n \prod_{p|n} (1 - 1/p)$. Its numerical factors show why this function requires the integer labels and their sizes. No step in this arithmetic transport identifies a real embedding or the analytic scalar shape from an unlabelled monoid.

Theorem 8.4 (Numerical Euler products and the full multiset inverse). *With the numerical positive-integer labels and their usual multiplication,*

$$\zeta(s) = \sum_{n \geq 1} n^{-s} = \prod_{p \text{ prime}} (1 - p^{-s})^{-1}, \quad \operatorname{Re} s > 1.$$

If a supplied nonnegative self-adjoint operator A has a complete simple eigenbasis with eigenvalues $0, 1, 2, \dots$, then on that same domain $\operatorname{Tr}(1 + A)^{-s} = \zeta(s)$.

Conversely, let a multiset of real generators $q > 1$ have positive integer multiplicities m_q . Suppose its Euler product

$$Z(s) = \prod_q (1 - q^{-s})^{-m_q}$$

converges to a finite positive value for all sufficiently large real s . The full real function on any such terminal ray uniquely determines the numerical multiset, including multiplicities. For a nonempty multiset its least generator q_0 and multiplicity m_0 are recovered by

$$q_0 = \lim_{s \rightarrow \infty} (\log Z(s))^{-1/s}, \quad m_0 = \lim_{s \rightarrow \infty} q_0^s \log Z(s).$$

Removing the entire factor $(1 - q_0^{-s})^{-m_0}$ and repeating recovers every remaining generator. The empty multiset is exactly $Z \equiv 1$.

Proof. Expand the geometric factors over any finite set of primes. Unique factorization gives the sum over exactly the integers supported on those primes. Absolute convergence of $\sum_{n \geq 1} n^{-\operatorname{Re} s}$ permits the limit over all primes and proves the identity. In the indicated eigenbasis the operator $(1 + A)^{-s}$ is diagonal with entries $(n + 1)^{-s}$ and singular values $(n + 1)^{-\operatorname{Re} s}$. It is trace class precisely for $\operatorname{Re} s > 1$, with the asserted trace.

For the inverse choose a sufficiently large $s_0 > 0$ at which the product converges. Positivity of its factors gives

$$\log Z(s_0) = \sum_q m_q [-\log(1 - q^{-s_0})] = \sum_q m_q \sum_{k \geq 1} \frac{q^{-ks_0}}{k} < \infty.$$

Tonelli's theorem justifies the expansion. For each finite $M > 1$ every generator $q \leq M$ contributes at least M^{-s_0} , counted with multiplicity. There are therefore only finitely many such generators. In particular the multiset is locally finite down to one, is countable, and if nonempty has a least $q_0 > 1$ of finite multiplicity m_0 .

For $s \geq s_0$ multiply the logarithmic series by q_0^s . Every term is $(m_q/k)(q_0/q^k)^s$ and is bounded by its value at s_0 . Those bounds sum to $q_0^{s_0} \log Z(s_0) < \infty$. The ratio q_0/q^k equals one exactly when $q = q_0, k = 1$, and is otherwise strictly less than one. Dominated convergence consequently gives

$$q_0^s \log Z(s) \longrightarrow m_0, \quad \log Z(s) = m_0 q_0^{-s} (1 + o(1)).$$

Taking the negative $1/s$ power gives the first limit and the second is already proved. Divide $Z(s)$ by the entire recovered factor, or equivalently subtract $-m_0 \log(1 - q_0^{-s})$ from its logarithm. The remaining product still satisfies the same convergence hypothesis. This removes all powers of q_0 at once, including those that coincide numerically with another generator. Repeating is therefore valid even in the presence of such coincidences. Every specified generator has only finitely many predecessors by the bounded-interval argument, so the iteration

reaches all of them. If none remains the logarithm is zero; if one remains it is strictly positive. This proves both uniqueness and the empty-product assertion. $\square$

The inverse in Theorem 8.4 retains the positive Euler-factor category and the numerical exponential coordinate q^{-s} . It does not reconstruct an addition law from abstract factors or resolve any collision in Theorem 8.3. Across these sections, every construction and converse has its mathematical proof and its retained foundations stated explicitly. The series, topology, matrix, stochastic, and arithmetic development is included in the complete Lean 4 formalization with no admitted proofs.

9 Global reconstruction closure and irredundancy

The complete identifying presentations below reconstruct the same mathematical object Σ exactly. Their different forms do not make them intrinsically distinct mathematical objects. The global reconstruction theorem retains each presentation's candidate class, calibrations, and any specified context.

Definition 9.1 (Identification and a retained context). An identifying presentation consists of a candidate class, its marked observation map, the specified observed value, and a proved inverse on that class. A reconstruction recovers the object whose observation was taken. A canonical construction in a context C is a specified map F_C giving a presentation of Σ in C ; it identifies an arbitrary supplied object only if that object satisfies the construction's identifying conditions. Contexts include the coordinate, ambient category and any designated argument. The reconstruction of an observed object and the identification of an arbitrary supplied object therefore have distinct quantifier structures.

Let E_Σ be the following finite collection of complete presentations, with every candidate class and calibration stated in the referenced theorem retained. Grouping presentations in a table row does not discard their distinct observation maps.

Presentation family	Complete identifying data	Results
Intrinsic entries	H , I , or the positive normalized p , in the marked coordinate	1.2
Differential core	Calibrated curvature, Riccati/exact flow, recentering	2.1–2.2
Group core	Normalized logarithm, calibrated cocycle, typed completion	2.3, 2.4

Presentation family	Complete identifying data	Results
Convex/projective core	Anchored Bregman/symmetric slice, exact conjugate, calibrated self-concordance/Hessian metric, projective/Schwarzian data	2.6–2.9
Gamma transforms	Complete law, Laplace, moments, Mellin line, characteristic function; entropy-maximizing law	3.2, 3.21
Poisson	Normalized recurrence, centered CGF, exact oriented KL slice	3.5
Gumbel/Haar	Two marked max identities and anchor, or complete marked Haar-weighted density	3.7, 3.8
Linked deficit	Both prescribed observations: deficit law and coordinate involution, linked to the same candidate potential	3.11
Other exact probability nodes	Normalized survival/logistic/Green; complete Stieltjes data; atom-free size-bias/equilibrium inverse	3.13, 3.15, 3.16
Formal/analytic series	Todd unit tower with constant one; complete Todd, A-hat, L and marked normalized χ_y , $y \neq -1$	5.1–5.3
Tree inverse	Complete marked rooted labelled EGF/Borel PGF and its analytic inverse germ	3.18
Marked rational presentation	Calkin–Wilf/Farey numerical labels and the specified full-domain Möbius extension	8.1–8.2
Measure/operator observations	Full-line Stein identity; linked mixing measure samples; complete marked polynomial orthogonality; full integer tail in the Bernstein class	4.1, 4.15, 4.8, 4.16

Formal/local nodes identify their formal germs directly. Identifying a supplied global function from a germ retains the connected real-analytic candidate class. The explicit expression also constructs its canonical global representative. Neither assertion identifies an arbitrary smooth continuation after that class has been forgotten.

Theorem 9.2 (Intrinsic uniqueness and reconstruction closure). *There exists a unique calibrated intrinsic analytic object Σ on the marked positive coordinate represented by every complete identifying presentation in E_Σ . Every presentation in E_Σ reconstructs Σ uniquely within its stated candidate class, and Σ reconstructs that presentation exactly.*

Hence the complete presentations in E_Σ are not intrinsically distinct mathematical objects: they are exact presentations of the same intrinsic mathematical object Σ . In particular, every two presentations $A, B \in E_\Sigma$ are mutually reconstructible through Σ .

Proof. Existence is established by the explicit constructions in the referenced theorems, and their identifying inverses establish uniqueness in the corresponding candidate classes. In the formal cases, the retained continuation class is exactly that of Proposition 5.3.

For each $A \in E_\Sigma$, write

$$e_A : \Sigma \longrightarrow A, \quad d_A : A \longrightarrow \Sigma$$

for its encoding and identifying inverse. On their stated solution classes,

$$d_A e_A = \text{id}_\Sigma, \quad e_A d_A = \text{id}_A.$$

For $A, B \in E_\Sigma$, the reconstruction from A to B is

$$e_B d_A : A \longrightarrow B,$$

with inverse

$$e_A d_B : B \longrightarrow A.$$

Indeed,

$$e_A d_B e_B d_A = e_A d_A = \text{id}_A,$$

and, exchanging A and B ,

$$e_B d_A e_A d_B = e_B d_B = \text{id}_B.$$

Thus this is one global reconstruction theorem. Pairwise equivalences are consequences of the reconstruction through Σ , not additional hypotheses. $\square$

Theorem 9.3 (Canonical realization fibres). *For each of the following supplied contexts, Σ canonically determines the stated object uniquely under the indicated identifying conditions:*

1. the probability-convolution completion with marked time one: $\mu_r = \text{Gamma}(2r, 1)$;
2. the normalized local conservative Sturm–Liouville context: the unique marked Laguerre self-adjoint closure;
3. the finite-dimensional spectral/scalar-block matrix context: $\Phi_n(X) = \text{tr } X - \log \det X - n$;
4. supplied isotropic $\mathbb{R}^4$ with $t = \|z\|^2/2$: the standard Gaussian radial realization;
5. the one-ancestor iid Galton–Watson context: the critical Poisson(1) offspring/tree law;

6. the supplied universal complex Thom/Chern-character context: the correction Td^{-1} .

The complete time-one, coordinate, rank-one, radial, total-progeny or universal-line data, respectively, recover the intrinsic object under the corresponding identifying conditions. These are equivalences in a fixed context, not identifications after forgetting that context.

Proof. The existence, uniqueness and reverse statements are precisely Theorems 3.22, 4.5, 6.1, 7.1, 3.19 and 5.4. Their proofs retain the indicated contextual data. In the last case only full universal line data identify the series, not a single finite-base evaluation. The operator spectrum and the fixed squared-resolvent mixing law likewise do not identify an unmarked stationary coordinate law, by the explicit isospectral countermodel in the operator section. $\square$

Theorem 9.4 (Relative irredundancy and its deletion witnesses). *Fix real analytic foundations and the following target convention: the output is the intrinsic shape, a placed function on a marked r coordinate if requested, and any requested externally selected realization packets. A packet includes its own type and its selected object, so that deleting it does not leave an ill-typed datum such as a bundle without its base. Canonical recipes are fixed, and different external packets have no unstated identifying map between them.*

The sufficient decomposition is one intrinsic certificate, the optional placement (μ, a) , and the requested external selection packets. The components (H, I, p) of the intrinsic decomposition of Σ are mutually reconstructible, with no free shape parameter after calibration. The two free placed-family scalar coordinates are exactly $\mu > 0$ and $0 < a < 1$.

For a deletion test, remove all alternative complete certificates of the same intrinsic shape; in particular do not retain a linked realization whose scalar restriction already identifies I . Relax the intrinsic candidate to anchored real-analytic strictly convex J with $\int_0^\infty e^{-1-J} = 1$. In this declared deletion universe the decomposition is blockwise irredundant whenever the target distinguishes the alternatives described in the proof. No assertion of shortest encoding across arbitrary languages is made.

Proof. Sufficiency is the closure theorem, the exact placed reconstruction, and the fixed-context construction theorem. For intrinsic irredundancy put $g(t) = (t-1)^4 e^{-t}$ and $h(t) = (t-1)^4 e^{-2t}$. They are linearly independent analytic functions vanishing to order four at one. For

$$M(\epsilon, \delta) = \int_0^\infty e^{-1-I(t)-\epsilon g(t)-\delta h(t)} dt$$

bounded perturbations justify differentiation under the integral, $M(0,0) = 1$, and $\partial_\delta M(0,0) = -\int p(t)h(t)dt < 0$. The implicit-function theorem gives $\delta(\epsilon)$ preserving mass. Since $t^2 g''$ and $t^2 h''$ are bounded, sufficiently small parameters preserve positive second derivative. The affine anchors and even unit curvature at one remain fixed, endpoint divergence is unchanged, and linear independence

gives a different shape for $\epsilon \neq 0$. Thus the weakened generic class is not already another complete certificate of Σ .

At fixed $a = 1/2$, take $\mu = 1$ and 2; at fixed $\mu = 1$, take $a = 1/3$ and $2/3$. The intrinsic data agree but the closure points or curvatures and placed functions differ. This proves both placement deletion witnesses.

For external packets, keep the whole scalar model and every other packet fixed. A coordinate relabelling changes a marked presentation without changing its underlying abstract object. Gamma shapes two and three have the same integer spectral package but distinct stationary coordinate laws. Uniform sphere angle and a fixed direction have the same radius; a Gamma subordinator and $X_r = rT$ have the same time-one law and different increments. Sample a Borel size and choose a path or a star to obtain distinct tree laws. Spectral matrix perturbations preserve the rank-one seed while changing the extension. The two complex lines on $\mathbb{RP}^2$ in Proposition 5.5 retain the same rational characteristic image and differ integrally. Permuting prime generators preserves the abstract multiplicative monoid and changes its numerical labels. Finally $\mathbb{R}^2$ and $\mathbb{R}^3$ separate spatial dimension; on fixed $\mathbb{R}^2$, squared norm and its square separate orthogonal composition, and squared norm and twice squared norm separate the remaining radius scale even inside the additive class. Each pair is an independent expansion of the same remaining decomposition data and changes its named target. $\square$

Corollary 9.5 (Boundary of scalar reconstruction). *The complete scalar component alone does not identify a placed coordinate choice, an arbitrary stationary/vector/process/tree/matrix realization, actual spaces or bundles and integral characteristic classes, numerical arithmetic labels on a bare carrier, a designated spatial observable, spatial dimension, or the radial cancellation rule. Under supplied global nonnegative orthogonal additivity in dimension at least two the spatial observable is $c\|x\|^2$. Under the supplied radial operator and residual cancellation one obtains $D \in \{1, 3\}$, and $D \geq 2$ selects three. These are exact boundaries of the stated scalar reconstruction; additional identifying observations may supply further inverses.*

Proof. The deletion witnesses in Theorem 9.4, together with Theorem 7.2 and Propositions 7.3–7.4, preserve the entire scalar component while varying the designated targets. A profile different from that of Σ also cancels the radial residual in dimension three. The auxiliary four-dimensional Gaussian remains fixed in these spatial expansions. $\square$

9.1 Exact reconstruction and formalization

Theorem 9.2 proves the global reconstruction of Σ across all complete identifying presentations in E_Σ . The typed presentation maps encode this exact reconstruction and its consequences. Ordinary spectral data retain unitary information, and canonical realization fibres retain their stated contexts. Forgetting a coordinate and then selecting a canonical realization does not provide an identifying

inverse for an arbitrary candidate. The countermodels above establish the exact boundaries of the proved reconstruction theorem.

The complete mathematical development has been formalized in Lean 4 with no admitted proofs. The formal statements preserve the stated candidate classes, domains, calibrations, retained data, and identifying conditions of the corresponding mathematical theorems.

10 Further Reconstructions

The global reconstruction theorem is proved; the presently known complete presentations of Σ need not be exhaustive. Additional complete mathematical presentations may remain to be discovered.

The criterion for inclusion is exact bidirectional reconstruction. Within its stated candidate class, an additional presentation is complete only when it provides an explicit reconstruction from Σ , an exact inverse recovering Σ , a uniqueness theorem, and boundary or counterexample data identifying which weaker information fails to reconstruct it. Visual or formal resemblance alone does not satisfy this criterion.

Every additional complete presentation must reconstruct the same Σ exactly and be reconstructed from Σ exactly within its stated candidate class.

Acknowledgments

I am deeply grateful to my wife, Malaithong Phommasone, who stood beside me throughout the years in which this work was developed. Her patience, strength, and unwavering support accompanied me through the long course of this work.

I am also deeply grateful to my family for their continued encouragement, understanding, and support throughout this journey. I am grateful as well to Péter Tímár and his wife, Rebeka, and to the friends who stood beside me along the way. Their friendship, encouragement, and support meant more than can be expressed here.

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